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Internal Medicine Block 1/2 - 2022

HIV male - emaciated, lost 15kg, oral thrush

1. according to WHO staging, place this man into the relevant stage (1)

Stage 1: Primary infection, asymptomatic, generalised lymphadenopathy

Stage 2: recurrent URTI, weight loss <10%, mucocutaneous manifestations, herpes zoster

Stage 3: Prolonged fever/diarrhoea, oral candidiasis, weight>10%, severe bacterial infection

Stage 4: AIDS-defining conditions (CMV, Pneumonia, TB, HIV related encephalopathy)

2. 3 opportunistic infections that present in a CD4 count < 100 cells/micro-litre (3)

Toxoplasmosis/cytomegalovirus/cryptococcosis (<200=pneumonitis carinii, mucocutaneous herpes, oesophageal candidiasis) <300=thrush

3. Does he qualify to start ART (1)

Yes, start irrespective of CD4 Count

4. What first line drug is used as a fixed-dose combination drug (1)

Atripla (efavirenz, emtricitabine, tenofovir)

Elderly Female - BMI 32, with type 2 Diabetes Mellitus experiences incontinence when she sneezes or coughs.

1. 3 types of incontinence (3)

Stress, urge, mixed, overflow, reflex

2. 2 risk factors for incontinence in the elderly (2)

DM, vaginal childbirth, obesity, cognitive decline, smoking, CVA

56yr old Male - presents with bringing up rust coloured sputum, cough, fever, right sided chest pain. He is confused, is hypotensive, centrally cyanosed, has a high respiratory rate (34 breaths per min), and a heart rate > 120bpm with the rate being described as regular and thready. On chest radiography there is opacification of the right middle and lower lobes.

1. What abnormality is being described on radiograph and what is the most likely diagnosis (2)

Consolidation, pneumococcal pneumonia

2. 2 signs from the case that this man is in respiratory failure (2)

Hypotension, tachypnoea, confusion, tachycardia

3. 2 additional signs of respiratory failure (2)

Restlessness, hypercapnia, central cyanosis, flapping tremor, bounding bulse, peripheral vasodilation

4. 4 complications of this disease (4)

ARDS, Respiratory failure, Death, Sepsis , Lung abscess

Women presents with weight gain, cold intolerance, brittle nails and hair after having surgery on her neck which she does not remember much about.

1. What is the most likely diagnosis (1)

Hypothyroidism

2. 2 causes of this condition (2)

Post-ablatio, iodide def, drugs(lithium)

3. What laboratory test would you do (1)

Thyroid function tests/thyroid antibodies (TSH, T4, T3)

4. Treatment of this patient (1)

Replace the deficiency (Levothyroxine=T4=Eltroxin)

Patient presents with acute onset of central chest pain radiating to the left arm. He is hypertensive and has a 20-year pack year. A diagnosis of ACS (acute coronary syndrome) is made.

1. List two other possible causes of the chest pain relative to this patient (2)

Myocardial ischemia, angina, lung cancer

2. 3 types of ACS (3)

STEMI, NSTEMI, Unstable angina

3. A non-fatal complication of this (1)

Non-fatal arrythmia

Youngish male presents with palmar erythema, spider naevi and jaundice.

1. What are 2 signs on general examination that you would see that would confirm cirrhosis (2)

Jaundice, clubbing, parotid glad enlargement, palmar erythema

2. This patient presents with caput medusae and splenomegaly. What are these findings in keeping with (1)

Portal hypertension

3. 2 signs that would indicate that this patient is in liver failure (2)

Hypoglycaemia, bleeding tendencies, jaundice, splenomegaly, rickets

Patient presents with dysphasia preventing solids and liquids.

1. Two non-neoplastic causes of this dysphasia (2)

Primary oesophageal disorder, scleroderma, achalasia

2. 3 investigations to be done (3)

Endoscopy, oesophageal PH Monitoring, barium oesophagram

Women presents with butterfly rash, fever, hair loss and describing her fingers turning white.

1. What phenomenon is being described by the women (1)

Raynaud's Phenomenon

2. What is the most likely diagnosis (1)

Systemic lupus erythematosus

3. What are 3 diagnostic criteria for this condition that are not mentioned in the clinical vignette (3)

Butterfly rash, Discoid rash, non-scar alopecia , Oral ulcers, Synovitis, Serositis, Proteinuria, Seizures/psychosis, Haemolytic anaemia

Young male, type 1 Diabetes mellitus, presents with dizziness and weakness

1. What emergency is being described (1)

Diabetic ketoacidosis

2. What are 2 precipitating factors for this (2)

Omission of insulin and underlying illness

3. What auto antibody would you see in this patient (1)

Insulin autoantibodies

4. What electrolyte abnormality would also be noted (1)

Hyperkalaemia

Male presenting with proprioceptive loss and glove and stocking syndrome.

1. What anatomical site is affected (1)

Upper and lower extremities

2. What other anatomical site is affected given the proprioceptive loss (1)

Cerebellum

3. What is a symptom of proprioceptive loss (1)

Balance issues, uncoordinated movement, clumsiness , poor postural control

4. What bedside test would you perform (1)

Romberg's Test

Women presenting with strabismus (left eye is medially rotated)

1. What abnormality is being described (answer would be strabismus) (1)

Strabismus

2. What cranial nerve palsy is seen (1)

CN 6 (abducens)

3. What would the cause be (1)

Stroke

4. What investigation would you do to confirm your suspicion (1)

MRI

Rheumatoid Arthritis question:

1) Foot deformities of RA (2)

Hallux valgus, Clawing / subluxation of toes, Separation of toes, Callosities, Collapse of arch

2) Antibodies of RA (2)

Rheumatoid Factor and Anticyclic Citrullinated Peptide Antibodies

3) First line drug of choice (1)

Methotrexate (DMARD)

EXTRA Question 5: list 4 classes of ARVs.

Nucleoside reverse transcriptase inhibitors, non-Nucleoside reverse transcriptase inhibitors, protease inhibitors, fusion inhibitors, integrase inhibitors, co-receptor antagonists

TB Trug regimen:

Rifampicin, isoniazid, ethambutol, pyrazimide

Block 3/4 Questions 2022:

Pleural effusion

1. Light's criteria
   Lights criteria:
   Serum protein>0,5
   Serum LDH>0,6
   Effusion LDH>2/3 of upper limit
   =exudate
2. Clinical presentation
   Pleuritic chest pain, ↓expansion, stony dullness, ↓vocal fremitus and resonance, Bronchial breathing above, tracheal deviation away
3. Differentials
   congestive heart failure, cancer, pneumonia, and pulmonary embolism.

Approach to paratracheal lymphadenopathy and hilar lymphadenopathy (most common lymphadenopathy)

1. Causes
   TB, sarcoidosis, lymphoma, histoplasmosis, and neoplasia
2. Hodgkin's vs non-Hodgkin's
   Presence of Reed-Sternberg cell, symmetrical, above diaphragm, few cell markers

   Stage I: A single lymph node area or single extra nodal site

   Stage II: 2 or more lymph node areas on the same side of the diaphragm

   Stage III: Lymph node areas on both sides of the diaphragm

   Stage IV: Disseminated or multiple involvement of the extra nodal organ

3. Diagnosis
   anaemia, lymphopenia, neutrophilia, or eosinophilia, Erythrocyte sedimentation rate, Lactate dehydrogenase, Serum creatinine, Alkaline phosphatase, A test for HIV, HISTOLOGY!

4. management
   Radiation therapy, Induction chemotherapy, Salvage chemotherapy, Hematopoietic stem cell transplantation

Mitral stenosis and atrial fibrillation in pregnancy (often a result from rheumatic fever)

1. Clinical presentation of both
   Mitral stenosis:
   Orthopnoea, paroxysmal nocturnal dyspnoea. Palpitations, chest pain, haemoptysis, thromboembolism when the left atrial volume is increased, ascites, oedema, and hepatomegaly (if right-side heart failure develops). Diastolic murmur
   AF:
   Absent P waves with irregular R-R Intervals
2. Association with pregnancy
   There is an increase in blood volume, heart rate and cardiac output in pregnancy (often as a result of Rheumatic fever) = iconic mitral valve
   ACE inhibitors, angiotensin-receptor blockers, direct renin-inhibitors, spironolactone, and eplerenone are contraindicated.
3. Management
   Diuretics, anticoagulants, B-Blockers, Ca channel blockers, anti-arrythmias
   Basically: Restore rate and restore rhythm

Chronic pancreatitis

1. Causes
   drugs, mumps, scorpions, autoimmune, hyperlipidaemia
2. Clinical signs on examination
   Epigastric pain relieved on sitting forward/hot water bottle
   Bloating
   Steatorrhea (fat)
   LOW
   Grey tuner's (umbilicus)
   Cullen's (flanks)

3. Investigations and findings

Elevated pancreatic enzymes
Hyperamylasuira
Raised Lipase is more specific
Ultrasound : enlarged
Gallstones often present
CT (calcification in chronic)
MRCP
Chronic: decreased faecal elastase

Portal hypertension

1. Presentation

Enlarged varices.
Enlarged spleen.
Gastrointestinal bleeding.
Ascites (fluid in your abdomen).
Signs of mild cognitive impairment.
Low blood cell counts.
Easy bleeding and bruising

2. Differentials

Cirrhosis, nodular regenerative hyperplasia of liver, schistosomiasis

Cirrhosis

1. Most common presentation

Portal hypertension

2. Presentation on hands

Clubbing, Dupuytren's contracture, palmar erythema

3. Management

General=good diet, avoid alcohol, NSAIDS, sedatives, opiates
Specific=hepatitis=high dose ursodeoxycholic
Wilson's = penicillin
Ascites = fluid and salt restriction = spironolactone
Spontaneous bacterial peritonitis = ciprofloxacin
Encephalopathy = lactulose

Gout

1. Most common locations

Big toe, ankle, foot

2. Risk factors

Obesity, alcohol, high protein, HPT, DM, dyslipidaemia

3. Management

Weight loss
Avoid: fasts, alcohol, aspirin, purine rich meat
NSAIDS
Urate lowering drugs (chronic)

Rheumatoid Arthritis

1. Predisposing factors

Twins, Human leucocyte antigen , smoking, hormones

2. Common locations

Hands and feet, worse in morning

3. Investigations

Rheumatoid factor, anticyclic citrullinated peptide antibody

4. Xray findings

soft tissue damage, reduced joint space, erosion at joint margin, deformities

Syphilis

1. staging
   Primary:<90days, macule-papule-chancre
   Secondary:4-10weeks after chancre, rash, condyloma Lata
   Tertiary= 20-40years Gumma
2. transmission
   skin to skin
3. management
   IM benzidine penicillin

Cranial nerves – myasthenia gravis

1. Signs
   Slowly increasing muscle fatigue, myasthenic snarl, ptosis, diplopia
   Curtain sign (raising one eye drops other)
   Peek sign + bells phenomenon (upward)
2. Investigations
   Antibodies:
   Anti-AchR/MUSK(muscle specific tyrosine kinase)
   EMG (nerve Jitter)
   CT
3. Co-morbidities
   hyperthyroidism, SLE, rheumatoid arthritis, scleroderma, ulcerative colitis, Addison disease, pernicious anaemia, red cell aplasia, Sjogren's syndrome, and sarcoidosis.

Diabetic ketoacidosis

1. DKA VS HH
   Insulin deficiency , glucagon excess and stress hormones
2. Diagnosis
   Hyperglycaemia and ketonuria (acetoacetate)
3. Management
   Fluids, insulin, potassium, bicarbonate
   Goal:
   Plasma glucose <11.1
   Serum bicarb>18
   Venous blood>7,3

Graves' disease

1. Risk factors
   Vit D deficiency/TSH receptor antibodies
2. Signs
   LOW, weakness, heat intolerance, alopecia, Plummer's nails
   Bi-lobe goitre with bruit
   Cervical lymph, vitiligo clubbing, bulging eyes
   Thyroid acropachy/dermopathy
3. Investigations
   TSH, T3, T3
4. Management
   T4 = Levothyroxine 1.6microgram / kg

Nephrotic syndrome

1. Risk factors
   Minimal change disease
   Diabetic nephropathy
2. Nephrotic vs Nephritic
   Proteinuria >3,5g
   Hypoalbuminemia
   Oedema (periorbital)

   Nephritic:
   Haematuria
   Hypertension
   Oliguria
   COLA URINE

3. Management
   Reduce oedema
   Treat underlying cause
   Reduce proteinuria
   Treat complications

Chronic kidney disease

1. Presentation
   Early = asymptomatic Fluid overload, anorexia, weakness, nausea, vomiting, pruritis, dark foamy urine
   Raised GFR, glomerular damage, scarring
2. Risk factors
   BP treatment, NSAIDS are causes
   HPT, DM, HIV
3. Management
   Hypertensive control
   Glucose control
   Lifestyle modification
   Treat complications
4. Criteria for renal replacement therapy
   Persistent Hyperkalaemia
   Pulmonary oedema
   Intractable acidosis
   Oliguria
   Uraemic pericarditis

Internal Medicine OSCE block 5 2022

Question 1:

An old man with a significant smoke pack history, presented with cough and dyspnoea. On Xray, a large mass was located on his right lung.

1. List 4 physical findings on examination:
   Clubbing, wasting, pallor, lymphadenopathy
2. What is the most likely diagnosis?
   Carcinoma of the lung (probably bronchial carcinoma)

Patient presented with an acute onset of rhinorrhoea, shortness of breath, sore throat, and a cough.

1. What is the likely diagnosis ?
   Bronchitis
2. List 3 viruses that can cause the above disease:
   Corona virus, rhinovirus, Influenza A and B, parainfluenza, adenovirus

Question 2:

Patient presented with orthopnoea, with a loud S3 and a pansystolic murmur heard at the apex

1. Grading of orthopnoea:
   Graded by the number of pillows used by the patient
2. 3 signs of heart failure:
   Peripheral oedema, ascites, wheeze, cold peripheries, anorexia, facial engorgement, raised JVP
3. Most likely diagnosis ?
   Mitral regurgitation

Patient with an ejection systolic murmur heard at the aortic area with a click, presents with dyspnoea and dizziness.

1. Where is the aortic area located?
   2/3rd intercostal space at the right sternal border
2. Diagnosis?
   Aortic stenosis
3. What is the character at the apex?
   Heaving
4. What does the above apex character mean?
   Pleasure loaded
5. Complications of the above disease?
   Heart failure, stroke, blood clots, arrythmias, death

Question 3:

Patient with a heartburn associated with meals and a retrosternal pain

1. Diagnosis?
   GERD
2. Other conditions that cause this type of pain?
   Angina, MI, pericarditis, atrial fibrillation, pulmonary embolism
3. 2 non-pharmacological managements?
   Weight loss, smoking cessation, avoid spice/hot food

Patient had MI; his father had a history of a MI at 20 years old. Both parents were on medication for high cholesterol levels. Patient also has depositions behind his tendons

1. Name the greater condition that the boy has?
   Dyslipidaemia
2. Name the first line drug used for this condition?
   Statins
3. Side effects of this drug?
   Headache, constipation, diarrhoea, flatulence
4. What are the depositions called?
   Xanthoma (xanthelasma is by the eyes)

Question 4:

Patient admitted after binge drinking. [resented with ataxia. Blood levels and CT scan are all normal

1. Diagnosis?
   Delirium
2. The minimum and maximum GCS scores?
   3-15
3. What vitamin would you administer?
   Vitamin B12
4. What are the side effects of said vitamin deficiency?
   Anaemia, fatigue, poor balance, memory loss, reduced sensory sensation

A 77-year-old man presents with tremors at rest for the past 12 years

1. Diagnosis?
   Parkinson's
2. 2 Other expected findings?
   Bradykinesia and hypertonia
3. Is the family history important? Why?
   Yes, Parkinson's is hereditary
4. What brain imaging would you use?
   MRI and Functional neuroimaging

Question 5:

Patient presented with a GFR of 29

1. What chronic kidney disease stage is this?
   Stage 4
2. 3 signs of a patient with CKD:
   Fluid overload, foamy urine, nocturia, polyuria, weakness, bruising
3. Cause of proteinuria in this patient?
   Chronic kidney disease (increased permeability of glomerular wall and impaired reabsorption)

Diabetic patient presents with chronic kidney failure and paraesthesia

1. Cause of the paraesthesia?
   Diabetic neuropathy
2. Cause of proteinuria in this patient?
   Diabetic nephropathy
3. Drug for proteinuria?
   ACE-inhibitor
4. 2 macrovascular complications of this diagnosis?
   Ischaemic heart disease, CVA

Question 6:

Patient with a positive rheumatoid factor and diagnosed with rheumatoid arthritis

1. 2 extra-articular manifestations of RA in the lungs?
   Interstitial lung disease, pleural effusion
2. 3 signs in the hands of this disease?
   Swan neck deformity, boutonnieres, ulnar deviation, z deformity of thumb

Patient with morning stiffness that lasts less than 20minutes with fatigue

1. Diagnosis?
   Osteoarthritis
2. Underlying pathology?
   Degenerative joint disease where the flexible tissue at the end of bones breaks down. Hyaline articular cartilage is the primary site
3. 2 non-pharmacological managements?
   Exercise and weight loss

Question 7:

IV drug user presented with signs of lung pathology, and isolated organism found was Staph Aureus

1. Explain the pathophysiology?
   (IE Or lung abscess)
2. List 2 investigations that would assist you with your diagnosis.
   CXR, Echo

HIV positive patient was diagnosed with cryptococcus

1. What is one investigation you would do?
   CSF and CT
2. 2 drugs that can treat?
   Bactrim, fluconazole
3. 2 HIV drug classes listed as first line treatment?
   2 NRTI and one NNRTI

Question 8:

Patient presents with bloody diarrhoea

1. What 2 organism caused bloody diarrhoea
   Shigella, salmonella, E. Coli
2. What are 2 stool samples that you would do
   Stool MC&S and FOBT

Internal medicine OSCE Block 1 and 2 2021

Question 1:

Hypertensive and diabetic patient presents with central crushing chest pain for the past 2 weeks. Pulse was rapid and BP was low.

1. Name 3 life threatening causes of this chest pain
   MI, aortic dissection, pericarditis
2. Interpret the ECG showing STEMI (identify 2 pathological features in this ECG)
   ST elevation, inverted T Waves
3. What definitive treatment would you give this patient
   Anticoagulation therapy
4. Name 2 antiplatelet drugs
   Aspirin, Clopidogrel

Question 2:

45-year-old normal man had a 30mmol on random glucose tolerance test.

1. Name the type of diabetes
   Type 2
2. 3 microvascular complications of DM
   Diabetic nephropathy/neuropathy/retinopathy
3. The drug of choice
   Metformin
4. Side effects of the above drug
   GI symptoms: nausea, vomiting
5. What is the test used to monitor the long-term response to treatment
   HbA1C

Question 3:

Patient presents with chest pain for about a year with mostly dry cough. He has no fever, weight loss or night sweats. On examination he had a hyperinflated chest and a wheeze was present.

1. Most likely diagnosis
   COPD
2. 3 risk factors for this condition
   Smoking, air pollution, second hand smoking, genetics
3. What 4 test would you do to further investigate this patient's condition
   CXR, CT, ABG, Spirometry
4. 2 physical signs seen with a hyperinflated chest
   Barrel chest, impalpable apex, hoovers sign

Question 4:

A young patient of 13 years presents with a sudden onset of right sided hemiplegia plus aphasia. Apex was palpated at the 5th ICS midclavicular line and a pulse deficit was present.

1. Where is the anatomical lesion
   Left cerebral cortex (likely MCA)
2. Pathogenesis of the hemiplegia
   Brain damage localised to the cerebral hemisphere opposite to the affected side
3. 3 causes of hemiplegia considering the patient's age
   Cardiac disease, congenital heart disease, acquired heart disease
4. 3 possible CVS findings in this patient
   Irregularly irregular pulse, palpitations, chest pain, absent p waves, irregular R intervals

Question 5:

Patient was diagnosed with a gastric ulcer.

1. Name 4 risk factors
   Smoking, alcohol abuse, hot/spicy food, stress
2. 2 treatment options
   Proton pump Inhibitors, H2 Blocker
3. 3 serious complications
   Bleeding, penetration, perforation
4. Why would you biopsy the edge of the lesion?
   Less likely to cause perforation

Question 6:

Young patient with periorbital oedema and bilateral pitting lower leg oedema. Her urine was frothy with 3+ protein.

1. Differential diagnosis
   Nephrotic syndrome
2. Name 2 methods of testing for proteinuria
   Urine analysis and U&E
3. Other impaired laboratory findings
   Haematuria, glycosuria, nitrates
4. Which test would give you a definite diagnosis
   Urinalysis on microscopy
5. What causes this disease
   Minimal change disease, diabetic nephropathy
6. 3 drugs that can be used as treatment
   ACE-I/ARB/Diuretics

Question 7:

A young female of 24 years presents with lethargy and pallor with the following FBC results: Hg was low, WCC was normal, MCV was low.

1. Differential diagnosis
   Iron deficiency anaemia
2. What 2 questions would you ask on history?
   Diet and menorrhagia
3. Name 2 extra tests that you would do
   Iron studies and smear
4. Name 2 indices from the findings that you were given that resulted to the conclusion of your diagnosis
   Hg was low, MCV was low, pallor, lethargic
5. Name 3 signs on exams to support your differential
   Koilonychia, angular cheilosis, atrophic glossitis

Question 8:

Young patient presents with a tinge of jaundice and acute onset of nausea and vomiting. Abdomen is non tender plus patient has 2cm tender hepatomegaly. No signs of chronic liver disease. (LFTs, Hep A and Hep B markers were given plus serology.

1. Likely diagnosis
   Acute viral hepatitis
2. What is the likely cause of your diagnosis
   Hepatitis A
3. Route of transmission for this disease
   Faeco-oral
4. More commonly found in which patients
   Travelers, living in crowded areas, poor sanitation, endemic to Africa and south America
5. What management principle would you use on this patient
   Supportive management -> metoclopramide for nausea and vomiting with non-hepatotoxic analgesia

Internal Medicine Block 3 & 4 2021

The following topics and questions were tested:

MEQ

Diarrhoea

1. GI causes of clubbing
   IBS, celiac disease, ascariasis, lymphoma of GIT, amoebiasis
2. What organisms cause bloody diarrhoea(dysentery)?
   E coli, Shigella, salmonella, entamoeba histolytica
3. Which endoscopic investigation would you do if mucous is present in diarrhoea?(Crohn's disease, ulcerative colitis, IBS)
   Colonoscopy

Peptic ulcer disease

1. Risk factors
   Smoking, alcoholism, stress, spicy foods
2. What class of drug is used for treatment?
   PPI or H2 blocker
3. Complications of peptic ulcer disease?
   Bleeding, penetration, perforation, gastric outlet obstruction
4. Which endoscopic modality would you use to prevent rebleed of ulcer?
   Endoscopic haemostasis

Seizures

1. Causes of seizures in HIV patients (name 2 aetiologies)
   Mass lesion, meningitis, HIV-encephalopathy
2. Signs of raised ICP on fundoscopy
   Papilledema, ocular venous engorgement, blurring, haemorrhage over optic nerve disk
3. Contraindications for LP
   Raised ICP, skin infection at site, trauma to lumbar vertebra

Stroke

1. Types of strokes
   Ischaemic vs haemorrhagic
2. 2 signs of LMN facial weakness
   Total paralysis of side of face, drooling, droop

Pleural effusion

1. Light's criteria
   Serum protein>0,5
   Serum LDH>0,6
   Effusion LDH>2/3
   =exudate
2. Causes of exudative effusion
   Infection, pneumonia, cancer, TB, Pulmonary embolism

Pneumonia

1. What are the features of respiratory distress?
   Tachypnoea, tachycardia, pale, agitation, increased effort(use of expiratory muscles)
2. What are the clinical signs of pneumonia?
   Pyrexia, cyanosis confusion, tachypnoea, tachycardia, hypotension, signs of consolidation, pleural rub, Crackles, resp distress
3. What investigations would you do?
   Oxygenation, bp, FBC. U&E, LFT, CRP
   CXR: Lobar infiltrates
   Sputum: GeneXPT, MCS
   TB, HIV, CD4
4. Organisms causing pneumonia
   Strep. Pneumoniae
   H. Influenza

Acute coronary syndrome - STEMI

1. Risk factors
   Smoking, hypertension, diabetes mellites, obesity
2. Emergency treatment for ACS
   (MONA B)
   Morphine, Oxygen, Nitrates, Anticoagulant, Beta-blocker
3. Complications of ACS
   (C PEAR DROP)
   Pericarditis, emboli, aneurysm, rupture, Dressler's, rupture, papillary ,muscle rupture
4. What are the clinical signs of heart failure?
   Tachycardia, tachypnoea, raised JVP, peripheral oedema, hepatomegaly

Osteoporosis

1. Risk factors
   Low oestrogen, anorexia, decreased vit d, smoking, alcohol
2. Investigations for diagnosis
   Dual energy Xray absorptiometry T-score (<2.5), Bone mineral density, serum calcium, PTH, phosphate
3. What treatment is given in post-menopausal women?
   Calcitonin, oestrogen therapy, bisphosphonates
4. Mechanism of osteoporosis in post-menopausal women
   Drop of oestrogen leads to more bone resorption that formation

Dementia

1. What screening exam is done?
   Mini mental state examination
2. Causes of dementia
   Alzheimer's or Lewy bodies, vascular issues also a possibility
3. Reversible causes of dementia
   Depression, infection, medication

MSK? CNS? – 19yo with progressive weakness after a short flu

1. Diagnosis
   Myositis
2. Name 2 other neuromuscular causes of respiratory failure
   Myasthenia gravis, lambert-Eaton syndrome, botulism
3. Bedside investigations to measure respiration
   Spirometry tests

Osteoarthritis

1. What are the features on radiography?
   LOSS:
   Loss of joint space, Osteophytes, Subarticular sclerosis, Subchondral cysts
2. What joints are affected in primary disease?
   DIP, MCP, knee, thumb

Rheumatoid arthritis

1. What antibody tests would you do?
   Rheumatoid factor, anticyclic citrullinated peptide antibodies
2. Forefoot changes in RA
   Bunions, claw toes, pain under ball, formation of calluses
3. What DMARD is used to treat it?
   Methotrexate

Malaria

1. What is the most common organism causing malaria in Africa?
   Plasmodium (anopheles' mosquito)
2. What is the vector for malaria?
   Anopheles mosquito
3. Complications of malaria
   Pulmonary oedema, acute renal failure, bleeding, severe anaemia
4. What laboratory investigations would you do?
   FBC, Blood smear, rapid diagnostic test (HRP2)
5. What is of note in the FBC results in malaria?
   Decreased HB, Thrombocytopaenia

Question bank: possible cases and questions

Rheumatology

Question 1: Rheumatoid arthritis

A 37-year-old female presents with a history of fatigue, fever, arthritis of her left and right MCP joints on both arms. She also has stiffness in her shoulders and knees especially in the morning for more than 1 hour.

1. What is the diagnosis ? (1)

Rheumatoid arthritis

2. Which joints are usually affected ? (2)

Small joints of hands and foot, the elbows can also be affected, usually spares the Distal interphalangeal joints

3. What are the hand deformities?(4)

Ulnar deviation of fingers, Volar subluxation of metacarpophalangeal joints. Z deformity of thumb, swan deformity of fingers, boutonnières deformity of fingers

4. What are the foot deformities? (4)

Lateral deviation of foot/ clawing of toes; collapse of arch; callosities; planovalgus deformity of foot(foot is not in linear plain, foot goes laterally); separation of toes, Crowding/ overriding of the toes(toes on top of each other), foot drop

5. What laboratory investigations/ activity markers would be needed to make a diagnosis?(3)

Rheumatoid factor, anti-CCP (cyclic citrullinated peptide) Anti-Nuclear Antibodies

6. What are 3 radiological changes in the condition? (3)‼

Soft tissue swelling, marginal erosions; joint space narrowing; juxta-articular osteoporosis; ankylosis; deformities of RA SS ME JAO

7. What is the treatment? (3)

NSAIDS, DMARDS, Analgesics; Biologicals (Anti-TNF agents)

8. What are two other syndromes that patients with this condition are likely to develop? (2)

Felty's syndrome, Sjögren's syndrome

9. What are some extra-articulation manifestations of (5) ‼

Systemic-fever; fatigue; weight-loss

MSK: Osteoporosis; muscle wasting; tenosynovitis; bursitis

Haematological: Haemolytic anaemia

Ocular: keratoconjunctivitis; Sicca; Episcleritis; scleritis; scleromalacia perforans

Pulmonary: Pleural effusions; Bronchiolitis obliterans; Caplan's syndrome

Sjögren's- dry red eyes; dryness of mouth and dental caries; parotidomegaly AND Felty's syndrome- Splenomegaly AND OTHER: Hepatomegaly; serositis (pericarditis, pleural effusion); vasculitis: digital vasculitis, ulcers, pyoderma gangrenosum

10. A patient with RA is more likely to have certain vascular conditions can you name 2? (2)

Myocardial infarction, Cerebrovascular accident

11. What are 3 factors that could result in the condition? (3)

HLA DR4 gene ; Smoking; periodontal infection; Obesity; Gut microbiomes

Question 2: SLE

a patient presents with butterfly rash, fever, hair loss and fingers turning white.

1. What is the phenomenon being described called? Describe the colour changes and what each means? (3) it is treated with warm water over hands

Raynaud's phenomenon , first fingers turn white due to lack of blood flow then blue due to vessels dilating to make blood flow towards them and finally red as blood flow returns

2. What is the likely diagnosis? (1)

SLE

3. What are 6 clinical criteria for this condition not mentioned in the clinical vignette ? (6)

Discoid rash, synovitis, serositis, photosensitive rash, proteinuria or red cell casts in urine, oral ulcers

4. What are 4 laboratory criteria ? (4)

+ANA, Anti-dsDNA, Low complement (C3, C4, C50), anti-phospholipid antibodies; anti-Smiths antibodies; +Direct Coombs test

5. How is lupus diagnosed ? I.e., how many criteria are needed? (2)

Lupus is diagnosed with >= 4 criteria being met (at least 1 clinical and at least 1 laboratory)

6. Tests for monitoring activity ? (3)

ANA, ESR, Complement

7. How is it treated? (3)‼

Sunblock, NSAIDS, Hydroxychloroquine

8. What other syndrome is related to 20-30% of lupus cases? (1)

Sjogren's syndrome

9. What does CLOTS stand for ? (5)

Coagulation defect, Livedo reticularis, Obstetric, Thrombocytopenia

10. How to check for the condition?(3)

Dry eyes, dry mouth of Sjogren's

11. What is a major complication? (1)

Renal failure, scaring, deformities

12. What drugs can cause lupus-like syndrome? (3)

Isoniazid, hydralazine, Procainamide

Question 3: Rheumatoid Arthritis

RHEUMATOID ARTHERITIS

1) What are some of the important factors which increase the risk of developing rheumatoid arthritis (RA)?

Twins, Human leucocyte antigen, Smoking, hormones

2) Which laboratory tests support a diagnosis of RA?

Rheumatoid factor, anticyclic citrullinated peptide antibodies

3) Which laboratory tests indicate disease activity in RA?

Increased CRP/ESR

4) What are the changes in the hands which occur in RA?

Swan neck and boutonnieres deformities

5) Which anatomical structures are affected in the joints of patients with RA?

Synovium

6) List the radiographic findings in RA.

Decreased joint space, soft tissue damage, erosion at margins, deformities

7) What are the clinical features of rotator cuff tendinitis?

Pain and swelling in the shoulder, limited motion, and weakness

8) What are the important domains used to classify patients with rheumatoid arthritis?

Inflammation, inflammatory markers, duration, serology

9) What Is the role of corticosteroids in RA?

Symptomatic relief

10) What are the common side effects of non-steroidal anti-inflammatory?

Nausea, vomiting, headache

11) What is the role of disease modifying anti-rheumatic drugs (DMARDS) in RA?

Reduce the progression of the disease

12) What are the common ocular manifestations of RA?

Keratoconjunctivitis, scleritis

13) List the features which support the presence of disease activity in RA?

Swelling, erythema, pain

14) What are the consequences of delayed DMARD therapy in RA?

Joint damage

Question 4: Osteoarthritis

OESTEOARTHIRITIS

A 65-year-old woman presents with severe knee pain and difficulty walking. Her symptoms are worse when mobilising and has progressed over the past one year. Radiographs of the knee joint reveal marked joint space narrowing and osteophytes.

1. What is the diagnosis ?

Osteoarthritis

2. What are the characteristic clinical features of this condition ?

Localised joint pain, with crepitus on movement, ache at rest, worse on movement

3. What are the radiographic features of this condition?

LOSS

Loss of joint space, Osteophytes, Subarticular sclerosis, Subchondral cysts

4. What are the risk factors ?

elderly/family/Collagen 2A gene/obesity

secondary due to Trauma, Perthe's diseases, acromegaly, inflammation

5. What are the non-pharmacological treatment modalities?

Loss of weight/exercise/joint replacement surgery

6. What are the mechanisms of joint pain in this condition?

The flexible tissue at the end of bones breaks down

A 71-year-old man complains of pain and swelling of the hand joints. He has early morning stiffness lasting approximately 20 minutes. On clinical examination he is found to have bony swelling of the distal interphalangeal and proximal interphalangeal joints.

1. What is the most likely diagnosis

Osteoarthritis

2. How do you classify this condition?

Kellgren and Lawrence system:

grade 0 (none): definite absence of x-ray changes of osteoarthritis

grade 1 (doubtful): doubtful joint space narrowing and possible osteophytic lipping

grade 2 (minimal): definite osteophytes and possible joint space narrowing

grade 3 (moderate): moderate multiple osteophytes, definite narrowing of joint space and some sclerosis and possible deformity of bone ends

grade 4 (severe): large osteophytes, marked narrowing of joint space, severe sclerosis, and definite deformity of bone ends

3. What is the relationship between osteoarthritis and ageing?

Inevitable consequence of growing old, described as a wear and tear disease

Question 5: Gout

GOUT:

A 56-year-old man presents with acute onset of pain and swelling of the right knee. He has hypertension and is on hydrochlorothiazide. His body mass index is 42kg/m2. The rheumatoid factor and anti-citrullinated peptide tests are negative

1. What is the differential diagnosis for his presentation?

Gout

2. What are the differentiating features of gout from other inflammatory arthritis on history and examination?

Obese, alcohol consumption, high protein diet

3. Which joints are usually affected in acute gout?

Big toe, ankle, foot

4. What are the predisposing factors for gout in this patient?

Age, obesity, hypertension

5. What are the trigger factors for acute attack?

Red meat, sugar drinks, alcohol

6. How will you confirm the diagnosis of gout?

Polarised light microscope, presence of uric acid crystals, serum urate

A 63-year-old woman with hypertension and diabetes mellitus presents with polyarticular joint pain and swelling for three months duration. She has small tophi at the MCP joints and bilateral elbows. The urea is 20mmol/L creatinine 230µmol/L.

7. Which drug is contraindicated in her acute management

hydrochlorothiazide

8. Why is not advisable to start the urate lowering therapy in this patient immediately?

Urate crystals can dislodge and cause an attack

9. When will you initiate urate lowering therapy?

Patients with more than 2 attacks per year. Uric urine crystals, nephropathy

10. Which urate lowering drug will you use in this patient.

Allopurinol

Question 6: SLE

A 23-year-old woman is diagnosed with SLE.

1. What are acute cutaneous manifestations of SLE?

Malar rash/ butterfly rash, which is erythematous over the cheeks, flat or raised, sparing the nasolabial folds. Heals with hyperpigmentation, Erythematous papule rash Photosensitivity on the sun exposed area, Mouth or nasopharyngeal ulcers which are usually painless

2. What are characteristics of arthritis in SLE?

Arthritis involves more than 2 peripheral joints with synovitis, typically non-erosive with reducible deformities (Jaccoud's arthropathy)

3. What features suggest lupus nephritis?

- Ranges from asymptomatic proteinuria to rapidly progressive lupus nephritis associated with end-stage renal disease.

- Presents with pedal oedema due to hypoalbuminemia, elevated blood pressure

- Urine analysis - dipstick may show proteinuria, haematuria, and casts (red blood cell/ white cell cast or granular cast)

- Urine protein creatinine ratio of > 0.05g/dL and 24hour urine protein > 1g/ 24 hours is considered significant.

- UPCR of > 0.3g/dl considered nephrotic range. It is important to exclude a urinary tract

4. What are the features of a haemolytic anaemia?

- Haemolytic anaemia

Maybe auto-antibody mediated or Coomb's positive

Clinical features are pallor or jaundice (due to high unconjugated bilirubin) Laboratory findings

Elevated lactate dehydrogenase and bilirubin,

Full blood count - the HB is low with elevated reticulocyte count Peripheral smear may show spherocytes and schistocytes.

5. What are the possible causes of pulmonary hypertension in SLE?

Pulmonary hypertension may develop secondary to a vasculitis and Interstitial lung disease, on history individuals may present with persisting dry cough and dyspnoea. Myocarditis and ischaemic heart disease may also occur in SLE

6. How do you assess disease activity in SLE?

Anti-Ds DNA, low complement and increased acute phase reactants (CRP and ESR) are features of disease activity.

Scleroderma

7. What are vasculopathy disorders seen in scleroderma

Raynaud's phenomenon

Telangiectasia: this are dilated capillaries typically blanching, occur on the skin of the face and chest, palmer surface of hands, oesophagus, and bowel

Pulmonary hypertension Scleroderma renal crisis (kidneys)

8. What is typical features of primary Raynaud's phenomenon?

1. Common in female, ages 15 to 30 years

2. No ischaemic changes

3. Nail fold capillary changes are normal

4. No autoantibodies (ANF negative).

9. What is the tissue hypoxia changes that can be seen in Raynaud's?

triphasic colour change. Initially pale or white due to vasoconstriction (vasospasm), then blue due to the deoxygenated blood on the peripheries and lastly red due sudden opening of the blood vessels (vasodilation). It is precipitated by cold, clinically the extremities fingers and toes are cold on examination

Question 7: Myositis

Myositis

10. What are the cardinal symptoms of idiopathic inflammatory myositis?

Proximal muscle weakness and pain, shoulder, and pelvic girdle. Difficulty getting up

11. What are some of the dermatological manifestations seen in dermatomyositis?

Heliotrope rash, Gottrons papules, V neck sign, shawl sign, holster sign, calcinosis and periungual erythema and nailfold telangiectasia

ant synthetase syndrome: mechanic hands.

12. What laboratory markers are used to diagnosed Inflammatory myositis

elevation in muscle enzymes (creatinine kinase, AST, and ALT)

Resp

Question 1: Pneumonia

A 26-year-old patient , HIV negative presents with Fever, dyspnoea, productive rust coloured cough and pleuritic chest pain. He does not smoke or has any previous illnesses. He has as RR of 27 breathes per min. He is confused and has high ICP when checked with a fundoscopy, he also has blue fingers.

1. What is the diagnosis?(1)

Pneumonia

2. What are differential diagnosis of the haemoptysis? (4)

Pneumonia, TB, Cancer, Acute respiratory tract infection, malignancy, COPD, Bronchiectasis

3. What are signs of respiratory distress? (4)

Use of accessory muscles, Tachypnoea, tachycardia, agitation

4. Is the patient in respiratory failure? (1)

Yes

5. Which type? What is the difference (3)

Type 2

T1: hypoxemic (prevents adequate oxygenation)

T2: hypoxia and hypercapnia (prevents excretion of Carbon dioxide)

6. Give features of Type 1. (4)

Dyspnoea, irritability, confusion, tachycardia, arrythmia, tachypnoea, cyanosis

7. Give features of Type 2. (5)

Headache, change of behaviour, coma, papilledema, warm extremities

8. 5 causes of a bounding pulse excluding respiratory failure. (5)

A BAD FAT CHAP

Aortic regurgitation, Beri-Beri, Anaemia, Ductus arteriosus, Fever, Av malformation, thyrotoxicosis, Cirrhosis, Hypercapnia, Arteriosus (truncus), Pregnancy

9. Explain the pathophysiology of the chest pain. (3)

inflammation-mediated disruptions in alveolar-capillary permeability, oedema formation, reduced alveolar clearance and collapse/decruitment, reduced compliance, increased pulmonary vascular resistance, and resulting gas exchange ...

10. What are signs on physical examination

Pyrexia, cyanosis confusion, tachypnoea, tachycardia, hypotension, signs of consolidation, pleural rub, Crackles, increased tactile vocal fremitus, decreased expansion

11. What are some non-respiratory causes of the dyspnoea? (5)

CARDIAC: Acute Left Ventricular Failure; Constrictive Pericarditis; Valvular heart Disease; Cardiomyopathy; Arteriovenous Shunting; ANAEMIA; Psychogenic; Massive ascites; Respiratory compensation of metabolic acidosis

12. What are the likely organisms causing the disease?(3)

CAP: Moraxella Catarrhalis; Haemophilus Influenzae; Streptococcus Pneumoniae.

13. What are organisms that cause hospital acquired- disease? (3)

Nosocomial: Staphylococcus aureus; Pseudomonas aeruginosa; Legionella spp

14. If the patient was immunocompromised, what are 3 OIs that could cause the disease?(3)

Immunocompromised: Cryptococcus neoformans; Pneumocystis Jirovecii; Mycobacterium Tuberculosis

15. What does CURB -65 stand for? (5)

Confusion <8 on MMSE: Urea >7mmol/l; Respiratory Rate>30 ;Blood pressure <90/60mmHg ;Age>65

16. What score will a patient need to be hospitalised? (1)

Hospitalised on CURB-65 score of>=2, however if suspicious can hospitalise a smaller score (children)

17. What investigations would be conducted? (5)

Air Blood Gas check pH, CO_2 and O_2 stats; Sputum -GeneXpert and MC&S; Blood: CRP; FBC; U&E/LFT? WHY? To evaluate for sepsis and end organ damage; legionella can also affect LFTs; Radiological: Chest X-ray

18. What could be seen radiologically?(3)

Areas of consolidation; pleural effusion; patchy reticular modular appearance ; ground glass appearance; cavitation

19. What are 5 complications of this condition (5)

Pleural effusion; Septicaemia; lung abscess, empyema; respiratory failure; hypotension, atrial fibrillation, pericarditis/ myocarditis; jaundice

Question 2: Bronchiectasis

A 34-year-old male presents with fever, dyspnoea, haemoptysis and coughing up purulent sputum. He had a recent PTB infection and completed his 6 months of treatment. He is HIV positive currently on treatment.

1. What is the diagnosis? (1)
   Post-TB bronchiectasis

2. What is the aetiology? (3)
   The patient had a previous TB infection which causes inflammatory damage ✅to medium and small airways resulting in permanent dilatation and thinning of these airways.✅ The airway is now susceptible to recurrent pulmonary infections which is precipitating the patient's condition.✅

3. What are 5 other causes of the condition besides the current infectious cause? (5)
   Inflammatory causes (SLE, RA, IBD) ;;; Congenital (primary ciliary dyskinesia; cystic fibrosis; Marfan's syndrome; Hypogammaglobinaemia);;; Endobronchial obstruction (Neoplasms, foreign bodies, lymph node compression);;; Other(inhalation exposure; allergic bronchi pulmonary aspergillosis; COPD);;; post-infectious (TB, pneumonia, whooping cough; HIV; measles)

4. What are the clinical findings not in scenario (4)
   General Exam: Clubbing
   Resp exam: Dec. Chest expansion over affected side; tracheal deviation; inspiratory course crackles +-Wheeze

5. What laboratory investigations should be carried out(specific to this patient) ? (3)
   - FBC-> infection
   - CRP-> infection and inflammation
   - Sputum -> GeneXpert + MC&S
   - HIV-> CD4 count; VL

6. What investigation is the gold diagnostic standard? (1)
   High resolution chest CT Why? Extent and distribution of the disease; Signet ring (wall thickening); Airway Dilatation

7. What will you see on X-ray? (3)‼‼
   Tram-track lines representing thickening of walls; cystic shadows; airway dilatation; tracheal deviation; possible air fluid level indicating cavitation

8. What other tests can be conducted and why ? (6)
   - Spirometry-> is there an obstructive pattern?
   - Bronchoscopy-> assess site of haemoptysis, exclude obstruction, obtain samples for culture✅
   - Serum immunoglobulins-> hypogammaglobulinemia✅
   - CF Sweat test-> CF✅
   - Allergic bronchopulmonary aspergillosis:✅
   - -Aspergillosis Precipitins
   - -Skin-prick test RAST
   - -Total IgE
   - SLE-> ANA, anti-dsDNA, complement (C3,C4,C50), antiphospholipid Ab
   - RA-> ANA, RF, anti-CCP✅

9. What are 3 complications ? (3)
   Recurrent pneumonia, pleural effusion; pneumothorax; haemoptysis; cerebral abscess , amyloidosis

10. Treatment? (5)
    Antibiotics; airway clearance techniques and mucolytics; corticosteroids; bronchodilators; surgery

Question 3: Asthma

A 25-year-old patient presents with an intermittent history of cough and chest tightness. On history you learn that the patient also has shortness of breath and a wheeze.

1. What is the diagnosis? (1)
   Asthma
2. List 4 other causes of a wheeze? (4)
   COPD, bronchitis, bronchiectasis, bronchiolitis, allergies, anaphylaxis
3. List 4 exacerbating factors for asthma. (4)
   Pollen, pets, cold weather, exercise, infection, emotions
4. How would you investigate this patient ? (3)
   Peak Expiratory Flow Rate if there is a >= 20% increase after Bronchodilator use
   Spirometry: it's obstructive if FEV1/FVC is <70%; it's reversible if FEV1 improved by >12% after BD
5. What is one drug that can be used to treat the patient ? (1)
   Bronchodilators- B2 agonist- Salbutamol // Corticosteroids
6. What dermatological lesions could the patient be predisposed to? (2)
   Atopic dermatitis; Psoriasis

Question 4: Pleural effusion

A patient presents with cough, dyspnoea, and chest pain. On examination they are SatO2 is 96 on room air. They have a stony dull area on the right lower zone. Bronchial breathing is present above this.

1. What is the diagnosis? (1)
   Pleural effusion
2. What are the investigations (5)
   Sputum MC&S and GeneXpert; Urinary lam; Urinalysis-> proteinuria, 24 hr urine collection; FBC -> anaemia and infection ;; CRP-> infection; LFT-> Cirrhosis; U&E -> AKI; autoimmune screen: SLE, RA; TFT-> hypothyroidism; Oestrogen levels -> ovarian tumour ; pleural tap-> send for MC&S, GeneXpert; Chemistry; Cytology; Chest X-ray-> look for Pleural effusion; Ultrasound Abdomen
3. What are light's criteria? (3)
   Lights criteria is a way of differentiating if a fluid is a transudate or exudate
   If Protein Fluid: Serum is >=0.5 then it's exudate
   If LDH fluid :Serum is >=0.6 then it's an exudate
   If pleural fluid LDH is >2/3 the upper limit of normal, then it's an exudate
4. What is a differential diagnosis if the fluid was an exudate? (5)
   Exudate: Resp: TB, pneumonia, pulmonary infarct, bronchogenic CA
   Serositis: SLE, RA
   Neoplasm: Malignant metastasis, lymphoma, mesothelioma; lymphangitis carcinomatosis
5. What is the differential diagnosis if the fluid was a transudate? (5)
   Transudate (low protein <25g/L): Inc. venous pressure: CCF; Constrictive pericarditis; fluid overload
   Hypoproteinaemia: Cirrhosis, Nephrotic syndrome; Malnutrition
   Endocrine: Hypothyroidism, Meigs syndrome (right sided effusion with ovarian tumour)
6. How do you manage the patient? (3)
   Pleurodesis; treat underlying cause; surgery if there are persistent collections and increased pleural thickenings on US

Question 5: COPD

A 45-year-old male presents with chronic cough for 5 months and breathlessness. On examination he has a barrel shaped chest and expiratory wheeze. His percussion note is hyperresonant. He smoked 3 packs a day for 10 years.

1. What is the likely diagnosis? (1)
   Chronic Obstructive Pulmonary Disease

2. What are risk factors? (3)
   Smoking, older >35; occupational exposure to smoke

3. What are some findings on clinical examination? (6)
   Dec. Chest rise; dec chest expansion; hoovers sign; tracheal tug; loss of cardiac and liver dullness/ hyper resonant percussion note; expiratory wheeze

4. What is the pathophysiology of a wheeze? (2)
   Wheeze is a musical expiratory sound that occurs when air moves out a turbulent airway due to blockage and bronchoconstriction of the airway.

5. What is his pack year ? (1)
   30 years

6. What tests would you conduct? (6) excluding Radiological
   ABG-> check if patient is in resp failure ✅;
   Lung function tests: spirometry FEV1/FVC <70% ✅; check reversibility if FEV1 improved more than 12% after Bronchodilator there is reversible ✅;;;; FBC-> anaemia & infection, Sputum GeneXpert &MC&S if suspect infection

7. What would you find on radiological testing? (3)
   Chest X-ray-> hyperlucent lung fields; blunting of costophrenic angle; more than 7ant ribs in MCL; flattening of hemidiaphragm

8. How would you manage non-pharm (3)
   Stop smoking; manage exacerbations; get vaccinated flu, covid, pneumococcus

9. How would you manage pharm… give stepwise approach (5)
   S1: SAB2A; S2: Add LABA; S3: Add Glucocorticosteroids; S4: Inc dose of corticosteroid; 5: Leukotriene receptor antagonist

10. What are 4 complications? (4)
    Cor Pulmonale; Respiratory failure; increased risk of lung Ca due to smoking history; spontaneous pneumothorax

Question 6: Interstitial Lung disease

A 67-year-old male presents with a dry cough, exertional dyspnoea, and fine basal crackles. He was a mine worker for 30 years.

1. What is the likely diagnosis? (1)
   Interstitial lung disease
2. What is the aetiology of this patient? (1)
   Pneumoconiosis- asbestosis; silicosis; berylliosis
3. What are other possible aetiologies due to a known cause? (4)
   Idiopathic; hypersensitive pneumonitis; Radiation pneumonitis; substance induced(Amiodarone, chemotherapeutic drugs; phenytoin; cocaine, heroin)
4. What are some aetiologies due to underlying disease? (4)
   Connective tissue disorders (SLE, RA, Scleroderma; dermatomyositis); Granulomatous diseases (sarcoidosis); infectious disease (TB; legionellosis)
5. What are crackles ? (2)
   Clicking, rattling, discontinued sounds
6. Is this a restrictive or obstructive disease? (1)
   Restrictive

Question 7: Lung abscess

A patient that was an IV drug user presents with cough, pleuritic chest pain and dyspnoea. He also has a productive cough with purulent foul-smelling sputum.

1. What is his diagnosis? (1)
   Lung abscess
2. What is the aetiology? (1)
   IV drug use leading to septicaemia or septic emboli
3. What are other possible aetiologies? (5)
   Inadequately treated pneumonia; aspiration (alcohol, oesophageal obstruction, bulbar palsy); bronchial obstruction (tumour, foreign body) pulmonary infarct; septic emboli(IV drug users, IE, septicaemia); subphrenic or hepatic abscesses
4. Signs on examination (4)
   Clubbing; LOW; dullness to percussion on affected side; Camphoric breath sounds over cavity
5. What are 3 laboratory investigations? (3)
   FBC ; ESR;CRP; Blood culture
6. What are 3 other investigations? (3)
   Chest X-ray-> cavitations with air-fluid level; CT chest to check for obstruction; bronchoscopy to look at extent and cause of haemoptysis and to take specimen samples
7. What is the general treatment ? (3)
   Antibiotics for specific cause; postural drainage; surgery

Question 8: Asthma

A 12-year-old male presents with shortness of breath and a wheeze. He usually has this at night often when he is near his dog. He also has eczema.

1. What is the most likely diagnosis?(1)
   Asthma

2. What is the pathogenesis? (4)
   There is a trigger that causes an inflammatory reaction in the airway ✅ this trigger is picked up as an antigen by Macrophages who recruit other cells to the site like neutrophils and lymphocytes and these release inflammatory cytokines (IL4 and IL5, TNF, leukotrienes) ✅ these cause an inflammatory reaction in the airway resulting in excess mucus production, hypertrophy/thickening, inflammation and swelling of walls of airway and bronchoconstriction. ✅ ✅

3. What are 5 triggers (5)
   Cold, exercise, emotion, dust mites, pollen, pets, mould, bugs, aspirin

4. Give 5 reasons why this would not be COPD? (5)
   The patient is less than 35 years old; there is no smoking history or occupational exposure; the patient gets a night-time attack, and the condition has a diurnal pattern; there is a history of atopy/ eczema; evidence of a trigger

5. How would you investigate this patient ? (2)
   PEFR >20% improvement & Spirometry FEV1/FVC <70% indicates OBSTRUCTION FEV1 improvement >12% indicates asthma

6. How would you classify his condition? What is it associated with? (2)
   Extrinsic (atopic) asthma due to presence of allergic nature ; associated with increased IgE levels.

7. How would you manage non-pharm? (3)
   Avoid triggers, aspirin and NSAIDS; stop smoking; reduce weight; gargle and rinse mouth after inhaler use; make sure patient understands treatment and technique

8. 3 risk factors for poor asthma outcomes? (3)
   Ever intubated for asthma; uncontrolled asthma symptoms; having more than 1 exacerbation in last 12 months; Low FEV1; incorrect inhaler technique/ poor adherence; smoking; obesity, pregnancy

9. What is the pharm treatment step wise? (5)
   S1 mild: Consider Low Dose ICS; as needed: SABA
   S2 moderate: LD ICS as needed SABA
   S3: LD ICS; LABA as needed SABA
   S4: Med High ICS; LABA as needed SABA
   S5 severe: refer

10. What is the normal SABA? (1)
    Salbutamol

11. What is the usual ICS? (1)
    Budesonide

Cardio

Question 1: Infective endocarditis

A 26-year-old female presents with a fever, anaemia, pan-systolic murmur, and frothy urine.

1. What is the diagnosis? (1)

Infective endocarditis

2. What is her valvular lesion?(1)

Mitral regurgitation

3. What are 3 features of the valvular lesion besides it being pan-systolic? (3)

A pansystolic murmur heard loudest over the mitral area, Radiation of the murmur to the axilla, loudest on expiration in the left lateral decubitus position

4. What 3 valves could be affected in order of most common to less (3)

mitral, aortic, tricuspid

5. Explain the pathophysiology of the condition and how it can lead to complications (4)

preparation of the cardiac valve for bacterial adherence, adhesion of circulating bacteria to the prepared valvular surface, and survival of the adherent bacteria on the surface, with propagation of the infected vegetation.

6. What immunological phenomena would be found ?(4)‼‼

glomerulonephritis, Osler nodes, Roth spots, or rheumatoid factor

7. What vascular phenomena would be found (4)‼‼

arterial emboli, septic pulmonary infarcts, mycotic aneurysm, intracranial haemorrhage, conjunctival haemorrhage, or Janeway lesions.

8. What is the 3rd phenomena?(1)

Microbiological phenomena

9. What criteria would need to be met to confirm diagnosis? (1)

Dukes' criteria

10. Elaborate on the major criteria. (2/3)

Major: 2 blood cultures, evidence of endocardial involvement,

11. Elaborate on minor criteria. (5)

Fever, IV drugs, Immune (Osler/Roth), Vascular (Janeway), Echo findings

12. How is it diagnosed? Is how many major and how many…

2 major/ 1 major and 2 minor/ or 5 minor

Question 2: Pericardial Tamponade

A 26-year-old male presents with a 3-day history of dyspnoea. His Bp is 80/42 mmHg, he has distended neck veins and his heard sounds are muffled.

1. What is the diagnosis? (1)
   Pericardial tamponade
2. What is the definition of the condition? (1)
   When there is a pericardial effusion with haemodynamic instability
3. What is the triad mentioned called? (1)
   Beck's triad
4. What is seen on ECG? (3)
   Sinus tachycardia; low voltage QRS complexes; electrical alternans (QRS complexes and T waves are at varying amplitudes)
5. How can the patient be managed? (1)
   Pericardiocentesis
6. What are 3 complication (3)
   Death, shock, heart failure

Question 3: Aortic Stenosis

A patient presents with exertional syncope, dyspnoea, and chest pain. On examination she has an ejection systolic murmur.

1. What is the diagnosis ? (1)
   Aortic stenosis
2. What are the possible causes? (3)
   Rheumatic fever; congenital defect-> younger patient's usually congenital bicuspid valve; degenerative defect -> particularly in elderly
3. What will the pulse be ?(1)
   An anacrotic pulse (slow rise and uptake) or late peaking and small volume; collapsing
4. What are 4 other signs on examination? (4)
   AS Apex beat is displaced, hyper dynamic ; May have a thrill over aortic area; S1 is normal; S2 is split.

Question 4: Acute Rheumatic fever

A young patient complains of a sore throat, chest pain that is better on leaning forward and subcutaneous nodules.

1. What is the diagnosis ?(1)
   Acute rheumatic fever
2. What is the pathogenesis of the condition?(3)
   A delayed immune response with group A beta- haemolytic streptococcus ✅, after 1-3 weeks antibody immune mediated damage ✅ can occur to heart valves, joints, basal ganglia, subcutaneous tissue ✅
3. What criteria is used to grade the condition ?(1)
   Modified Jones Criteria
4. What are the major criteria? (5)
   Migratory polyarthritis; Carditis; Subcutaneous Nodules; Erythema marginatum; Sydenham Chorea
5. What is the non-laboratory minor criteria? (4)
   Fever; arthralgia; previous rheumatic fever; prolonged PR interval
6. What is laboratory minor criteria? (3)
   CRP >30, ESR >60, leucocytosis
7. What is supporting evidence? (3)
   Evidence of Streptococcal infection: Inc. titre of anti-streptococcal antibodies ASO (anti-streptolysin O) positive throat swap for Group A beta-haemolytic streptococcus ;;;; recent scarlet fever
8. What investigations would you conduct?(5)
   FBC; CRP; ESR; ECG; Anti-Streptolysin O
9. How is a diagnosis made? (3)
   All need evidence of streptococcal infection; at least 2 major OR 1 major and 2 minor
10. What is the management? (4)
    Primary prevention(eradication of streptococcus)=Benzyl Penicillin G; anti-inflammatory treatment Aspirin for fever, arthralgia; Prednisone for Carditis ; supportive management _CCF- Digitalis, diuretics; Chorea-Diazepam; rest to joints ; secondary prevention
11. How long would you do secondary prevention ?(3)
    If no carditis then 5 years or till age of 21 which comes first; if ARF with carditis only then at least 10 years or well onto adulthood; if ARF with carditis and residual valvular heart disease, then for 10 years since last episode or until age of 40(sometimes lifelong prophylaxis)
12. What are complications of the disease ? (4)
    Valvular heart disease, infective endocarditis, Thromboembolic events (CVA, MI)

Question 5: Atrial Fibrillation

A 70-year-old patient with hypertension and on Enalapril presents with AF.

1. What would be 2 clinical features? (2)
   Irregularly irregular pulse; Pulse deficit; beat to beat S1 intensity variability
2. What are 2 RFs in this patient? (2)
   Elderly; hypertension
3. What are 4 causes of AF? (4)
   IHD; HEART FAILURE; HTN; PE; Mitral valve disease; Pneumonia; hyperthyroidism; caffeine; alcohol; post-op; hypokalaemia; hypo magnesium
4. Why would you anti-coagulate this patient? (2)
   AF leads to thrombus formation ✅ which can detach and cause MI, CVA so we anticoagulants to prevent clotting in heart and formation of thrombi ✅
5. 2 ECG findings (2)
   Absent P waves; irregular R wave intervals
6. Rate control (3)
   Rate-> BB ✅; rate limiting Ca2+ blockers ✅; Digoxin; Amiodarone ✅
7. Rhythm control (3)
   Rhythm-> Elective DC cardioversion; ✅Amiodarone if increased risk of cardioversion failure 4 weeks before; Amiodarone; Flecainide ✅ ✅

Question 6: Hypertension

A 45-year-old male presents with blurred vision, and dizziness. He has a BP of 150/94 mmHg on his previous visit; he now has a BP of 168/100mmHg.

1. What is the diagnosis? (1)
   High blood pressure ✅ NOT HYPERTENSION
2. Why can't you diagnose hypertension?(1)
   The patient is required to have 3 persistently high BPs at least 2 days apart
3. How would you manage this patient ?(3)
   Lifestyle modifications: Exercise; reduce salt and sugar; manage other co-morbidities; reduce stress
4. If the patient returned with a BP of 150/90 mmHg. What is the syndrome? (1)
   Hypertension
5. Grade his hypertension based on current reading (1)
   Grade 1B ; 140-159/90-99mmHg with >= 3 RFs; TOD or complications
6. 5 RFs for the above condition ?(5)
   DM; Hypercholesterolaemia; IHD in a first degree relative male45 female65; IHD; Obesity; lack of exercise; poor diet; Smoking ;Age; alcohol intake
7. Give 5 causes of secondary hypertension ?(5)
   RENAL: Renal artery stenosis; Chronic pyelonephritis; CT disease; Glomerulonephritis; polycystic disease ✅; ENDCRINE: CAH; acromegaly; Thyrotoxicosis; hypothyroidism; Conn's syndrome; Phaemocytochroma; Cushing syndrome ✅;OTHER Co-arc ration of aorta ✅; Drugs(Cocaine, Amphetamine, OCC); raised ICP, polycythaemia rubra Vera ✅ ✅
8. How would you differentiate using a bedside investigation ?(2)
   Blood pressure reading-> essential HTN will have a rise in diastolic blood pressure when patient is standing while in secondary there will be a FALL. ✅ ✅
9. What is another sign that can be present ?(1)
   S4-> due to atrial contraction and large volume of blood hitting a non-compliant ventricle. ✅
10. What are 3 micro vascular complications ?(3)
    Retinopathy; nephropathy; neuropathy
11. What are 3 macro vascular complications ? (3)
    IHD; MI; CVA; PVD
12. Give the grading of the fundoscopy exam and what each mean. (8)
    Grade 1- Sliver wiring of arteries only; Grade 2- AV nipping/nicking; Grade 3- Cotton wool spots, flame shaped haemorrhages; Grade 4 - papilledema
13. What is HTN emergency criteria ? (2)
    >=180/110 mmHg ✅with symptoms and/or accelerated TOD ✅
14. What drugs can be used to treat the patient? (3)
    Thiazide diuretics; ACE-I; CCB ; Spironolactone; Furosemide

GIT

Question 1: Ulcer

A 65-year-old patient presents with epigastric pain with nausea, bloating and burning sensation quickly after staring to eat a meal. She has anaemia and is anorexic as part of the ALARMS features. After investigations she is diagnosed with a duodenal ulcer. (15)

1. What is the symptom cluster being described by the patient known as ? (1)

Heart burn

2. What are 2 non-serious causes of the above symptom cluster? (2)

Pregnancy , stress, food, increase in progesterone

3. What are 3 other ALARMS features not mentioned in the clinical scenario? (3)

Loss of weight, melaena, recent onset, swallowing difficulty

4. What are 3 RFs for the condition? (3)

Alcohol abuse, NSAIDS, Tobacco, radiation therapy, H pylori infection

5. What are 2 other possible non-ulcerative differentials for this patient ? (2)❗

GERD, IBS

6. What is the best diagnostic investigation…

Endoscopy

7. What are the signs of alcoholism?

Blurred vision, loss of coordination, loss of speech, liver damage, vomiting, tachycardia, hypertension, Loss of weight

8. What are the signs of liver disease:

Hands: leukonychia, clubbing, palmar erythema, bruising, asterixis.

• Face: jaundice, scratch marks, spider naevi, fetor hepaticus.

• Chest: gynaecomastia, loss of body hair, spider naevi, bruising, pectoral muscle wasting.

• Abdomen: hepatosplenomegaly, ascites, signs of portal hypertension, testicular atrophy.

• Legs: oedema, muscle wasting, bruising.

Question 2: Haematemesis

A 24-year-old male presents with haematemesis. (20)

1. What are 5 specific questions you would ask the patient? (5)

When did it start? How much? What colour is it ? Do you have blood in stool? Before you had blood in vomit was there any period of intense retching? Do you have peptic ulcer disease? Do you have liver disease? Do you take NSAIDS, steroids? Do you smoke? Do you drink alcohol? Have you lost weight?

2. The patient has a suspected upper GI bleed, how can you tell this from the history? (1)

The blood in the vomit will be bright red

3. What is melaena stool? What GI bleed is it associated with? (2)

Malena stool is tarry coffee ground-stained stool that occurs due to an Upper GI bleed

4. The most likely cause is? (1)

Chronic peptic ulcer disease OR acute peptic ulcer (erosions)

5. What are 3 less common causes? (3)

Oesophageal or gastric varices; Mallory-Weis syndrome; Erosive or ulcerative oesophagitis; Gastric carcinoma; Amyloidosis

6. If the patient had a lower GI bleed what are some findings on history (2)

Coffee ground vomitus; haematochezia.

7. What are the 3 common cause: of a lower GI bleed ? (3)

Angiodysplasia; diverticular disease; colonic carcinoma; haemorrhoids or anal fissures

8. What are 3 less common causes? (3)

Massive GI bleed; IBD; Ischaemic colitis; Meckel's diverticulum; Small bowel disease (tumour, diverticula, intussusception)

9. What are 2 investigations we can conduct ? (2)‼‼

Upper GI endoscopy; colonoscopy; CT angiography; Meckel's scan

Question 3: Chronic liver disease

Young male 23yo, HIV + compliant on medication; presents with palmar erythema, abdominal pain, and jaundice for 8 months. He has a history of swimming in the lake near his house. He also drinks alcohol about 4 units every day. He has tattoos and has never had a blood transfusion before. He is of risky sexual habits. The patient is agitated.

1. What is broad diagnosis? (1)

Chronic Liver Disease

2. What is the differential diagnosis specific to this patient?(3)

Alcoholic liver disease, Schistosomiasis, Chronic Hepatitis B, or C

3. What are signs on general examination that would suggest cirrhosis?(5)

Palmar erythema; spider naevi; gynecomastia; petechia; jaundice; alopecia; clubbing; leukonychia, Terry's nails; conjunctival pallor; testicular atrophy

4. 3 signs of liver failure in this patient? (3)

Fetor hepaticus; asterixis; hepatic encephalopathy ; constructional apraxia

5. Grade his mental problem? (1)‼‼

Grade 2

6. What are the precipitants of hepatic encephalopathy? (4)

Gastrointestinal bleeding; infection; electrolyte imbalances(hypokalaemia, hyponatremia); hypoglycaemia; diuretic use/ dehydration, renal decompensation; alcohol use

7. 3 signs of alcoholism?(3)

Dupuytren's contractures; Fetor; parotidomegaly; fine resting tremor; staggering gait

8. Give 7 aetiology of the CLD? (7)

INFECTIOUS: Chronic Hepatitis B&C; Schioasomiasis; TB; Mononucleosis; TOXIC: Drugs: Anti-malaria; anti-TB; Paracetamol overdose; Alcohol; AUTOIMMUNE: Autoimmune hepatitis; Primary Biliary Cholangitis; Primary Sclerosing Cholangitis; LIFESTYLE: Non-alcoholic Fatty liver disease;;; GENETIC(Hemochromatosis; A-1 anti-trypsin deficiency; Wilson's disease);; CARDIAC: Right heart failure; Haemolytic anaemia; Chronic Budd-Chiari syndrome;;; NEOPLASTIC (Primary or secondary)

9. What are 5 investigations in this patient specifically?(5)

FBC-> check for infection; CRP, ESR-> infection and inflammation; LFT -> hepatic damage(Alanine Transaminase; aspartate transaminase) or cholestatic damage(gamma glutamyl transferase; alkaline phosphatase), albumin: checking synthetic function of liver, bilirubin-> markedly inc. in Alcohol despite small inc. in ALP; INR-> check function of clotting factors; Hepatitis Screen(Hep B:HBsAg; HBeAg; Anti- HBcAb; Anti-HBs Ab; Hep C: Anti-HCV Ab); Urine and Stool MC&S to check for Schistosomiasis; If all fail then biopsy

10. What are possible precipitants of the CLD? (2)‼

Infection, Alcohol use; intra-abdominal bleeding; electrolyte imbalances (hypokalaemia; alkalosis)

Question 4: Splenomegaly

A 24-year-old female patient presents with an overt Splenomegaly of 18 cm and caput medusae?

1. What are the features of Splenomegaly and caput medusae in keeping with? (1)

Portal hypertension

2. What other features could we expect to find in this patient ?(2)

Oesophageal, gastric varices(haematemesis); haemorrhoids (haemochezia); ascites

3. What is the normal size and location of a spleen? (2)

8-12 cm; located between the 9th and 11th ribs in the LUQ.

4. Does the patient have a massive spleen? Why/why not? (2)

No, a massive spleen would be >20 cm.

5. Give 4 features of a spleen?(4)

A spleen enlarged infero-medially; a spleen is usually not palpable; a spleen is not ballotable; cannot get above a spleen; has a splenic notch

6. Give some possible Pre-hepatic causes(3)

Splenic vein thrombosis; Neoplasm, hepatoma, pancreatic cancer; infection: appendicitis; diverticulitis; inflammatory: pancreatitis, IBD, sclerosing Cholangitis, Myeloproliferative disorders

7. Give some hepatic causes (3)

Schistosomiasis, cirrhosis(alcoholic hepatitis; fulminant hepatic failure); Veno-occlusive disease; modular regenerative hyperplasia

8. Give some post-hepatic causes (2)

Cardiac: Valvular HD(usually tricuspid regurgitation); Constrictive pericarditis; Right Heart Failure; Budd-Chiari syndrome

9. Other causes of Ascites and portal hypertension (3)‼‼

Other: Massive liver metastases, Hypothyroidism, Fatty liver disease of pregnancy, cardiac ascites

10. How would you investigate this patient ? (5)

SAAG; FBC; U&E; ESR; CRP; LFT; Albumin; INR; D dimer; US abdomen; CT abdomen; urinalysis; TFTs

Question 5: Acute pancreatitis

29-year-old John presents with epigastric pain that radiates to the back. He admits to binge drinking alcohol 12 hours before the onset of symptoms. His Bp was elevated, and his Heart rate was 112 BPM.

1. What is the most likely diagnosis? (1)
   Acute pancreatitis

2. Name 5 causes excluding alcohol (5)‼
   Gall stones, trauma, drugs(ARVs); hyperlipidaemia, post-ERCP, auto-immune, idiopathic, hypercalcaemia, genetic mutations

3. Explain why digestive enzymes are increased in serum? (2)
   Alcohol causes an inc. in toxic metabolites which causes spasm of the sphincter of Oddi: There is a blockage of excretion of pancreatic enzymes however synthesis still continues, this results in a high conc. of pancreatic enzymes in acinar cells, the enzymes then escape via the basolaminar membrane into the circulation causing an increase in pancreatic enzymes in the body

4. Name 2 diagnostic lab investigations (2)
   Serum amylase and serum lipase(more sensitive)

5. What are 2 x rays you would conduct and give 2 reasons each?(6)
   Abdominal X-ray: exclude other causes - bowel obstruction and bowel perforation & in severe disease: ileus of segment of small intestine("colon cut-off sign" Chest X-ray: elevation of hemidiaphragms, pleural effusions, basal atelectasis, pulmonary infiltration, ARDS,

6. What other radiological investigations? What would you look for in each (4)
   Abdominal ultrasound: look for gallstones, diffuse enlarged hypoechoic pancreas; CT abdomen: definitive diagnosis: assess severity, check for pseudocysts, also when patient not responding to initial conservative therapy

7. What are immediate complications?(2)
   Pancreatic pseudo cyst and pancreatic necrosis/ abscesses

8. What are long term complications (2)
   Pancreatic carcinoma, Diabetes mellitus

9. What is a lab test that predicts severe pancreatitis? What does it need to be?(2)‼
   CRP >150 mg/dL at 48 hrs

Question 6: Ascites

A 24-year-old female presents with overt abdominal distension. She is HIV positive. After examination she is found to have ascites.

1. What is ascites? (2)

Ascites is build-up of fluid ✅ in the abdominal cavity either due to portal hypertension or another peritoneal cause. ✅

2. What are 3 clinical examination you do to check for Ascites? (3)

Shifting dullness, fluid thrill, puddle sign

3. How would you investigate this patient? (5) Excluding SAAG

- Urinalysis-> check for proteinuria
- LABORATORY
- FBC-> anaemia, infection
- LFT-> Hepatocellular damage
- ESR-> inflammation
- CRP-> infection and inflammation
- U&E-> check kidney function, electrolyte imbalances
- Radiological
- Abdominal US-> confirms ascites, check for stones, masses, organomegaly, guides paracentesis
- CT abdomen

4. What does SAAG stand for? (1)

SAAG(serum ascites albumin gradient)

5. How do you make sense of the results with reference to cut offs? (4)

A SAAG > 11.1g/L indicates portal hypertension; a SAAG< 11.1 g/L indicates another cause

6. What other tests would you do on an Ascitic tap? (4)‼

Macroscopic appearance; MC&S; cytology; GeneXpert; ADA(adenosine deaminase)

7. If the patient had a SAAG of 13.6 g/dL, what other features could they have on examination(4)

Splenomegaly, caput medusae, haemorrhoids, haematemesis, melena stool

8. Give a differential diagnosis for Portal Hypertension?(8)

Pre-hepatic: Portal vein obstruction (tumours, portal vein thrombosis, pressure from adjacent lymph nodes, congenital portal vein stenosis), trauma ✅✅;;; Hepatic: Schisisomiasis ✅, veno-occlusive disease; cirrhosis: alcoholic hepatitis; fulminant hepatic failure; nodular regeneration hyperplasia, ✅::: Post-hepatic: Budd-Chiari syndrome ✅, RHF, constrictive pericarditis, valvular heart disease(tricuspid) ✅ OTHER: Fatty liver disease of pregnancy; myxoedema(hypothyroidism); massive liver metastasis ✅✅

9. Give a differential diagnosis for non-portal hypertension? (5)❗

Hypoalbuminemia due to malnutrition, CLD, nephrotic syndrome ✅;; pancreatic ascites/ biliary ascites ✅; serositis in CT disorders(SLE, RA) ✅;;;;;;;; PERITONEAL CAUSES——TB Abdomen, bacterial peritonitis ✅; malignancy- peritoneal carcinomatosis ✅

The patient suddenly develops a fever and becomes agitated.

10. What is the possible cause? (1)

Spontaneous bacterial peritonitis

11. What are 3 features on clinical examination?

Guarding, rigidity, rebound tenderness

12. What is the macroscopic appearance of Ascitic fluid? (1)

Turbid

13. What is the possible cause due to WCC? (2)

If neutrophils predominant: SBP; if lymphocytes predominant: TB peritonitis; Peritoneal carcinomatosis

14. The patient has neutrophil predominant. What are the likely pathogens ?(3)

E. coli; Klebsiella spp.; Streptococcus

15. What are 5 other addition tests can be done? (5)

The flowing:

- Glucose: low- infection, malignancy
- LDH: high- Malignancy, infection, or bowel perforation
- Amylase: high- pancreatic ascites or bowel perforation
- Bilirubin: high -bowel or biliary perforation
- Triglyceride: high in chylous ascites
- Cytology, CEA: suspected malignancy

Question 7: Hepatitis

Young patient present with a tinge of jaundice and acute onset of nausea and vomiting. Abdomen is non tender plus patient has 2cm tender hepatomegaly. No signs of chronic liver disease.

LFT: AST & ALP are rising then return to normal

IgG for Hep B: Raised (infers vaccination or chronic infection)

IgM for Hep A: rises on day 25 (implies recent infection)

1. Likely diagnosis?(1)
   Acute viral hepatitis

2. What the likely cause of your diagnosis(1)
   Hepatitis A

3. Route of transmission for this disease(1)
   Faeco-oral

4. More commonly found on which patients (4)
   Travellers, living in overcrowded areas; poor sanitation; endemic to Africa and South America

5. What management principle would you use on this patient.(2)
   Supportive management -> metoclopramide for nausea and vomiting; non hepatotoxic analgesia

Question 8: Nausea and vomiting

A woman presents with nausea and vomiting.

1. What are 3 non-serious causes? (3)
   Pregnancy ;alcohol use; drugs (digoxin, opiates, chemotherapy)

2. How would you treat symptomatically? (2)
   Fluid rehydration and Metoclopramide 10mg 8 hourly

3. What are 5 serious causes? (5)
   Gastrointestinal infections (food poising by Staphylococcus Aureus/ E.coli); small bowel obstruction; peptic ulcer disease with gastric outlet obstruction; motor disorders (gastroparesis from DM; or after gastric surgery); acute hepatobiliary disease; psychogenic(bulimia); raised ICP; renal failure

4. What are 5 laboratory investigations ? (5)
   FBC; U&E; LFT; CRP; glucose; amylase

5. Why would you want to do a ABG in this patient ? (1)
   To check for metabolic alkalosis due to loss of gastric content via vomiting

6. Other investigations ? (3)
   Plain abdominal X-ray; Upper GI endoscopy; Head CT

Question 9: Dysphagia

A 50-year-old patient presents with dysphagia affecting solids and liquids.

1. What is the possible DD?(4)
   Motility disorders: Achalasia; diffuse oesophageal spasm; Systemic sclerosis; neurological bulbar palsy

2. What would the DD be if they had solids then liquids? (4)
   Benign structure: Oesophageal web or ring; peptic stricture; malignant structure: pharyngeal cancer; oesophageal cancer; gastric cancer

3. The patient is found to have a bulbar palsy. What Cranial nerves are affected? (1)
   CN 9; 10; 11; 12

4. Differentiate a bulbar and pseudo bulbar palsy? (2)‼‼
   Bulbar is a LMN disease; Pseudobulbar is a UMN disease

5. What are signs of bulbar ?(4)
   LMN- Flaccid, fasciculation tongue; jaw jerk normal or absent; speech is quiet, hoarse, or nasally

6. What are signs of pseudobulbar?(4)
   UMN- Slow tongue movements; with slow deliberate speech; inc. jaw jerk; inc. pharyngeal and palatal reflexes: Gag reflex absent ; Weeping unprovoked by sorrow or mood-incongruent giggling ; difficulty chewing

7. Causes of bulbar (3)
   Bulbar: GBS; MND; Polio; Myasthenia Gravis; Syringobulbia; brainstem tumours; central pontine myelinolysis

8. Causes of pseudobulbar (3)
   Pseudo bulbar: Cerebrovascular accident; Multiple Sclerosis; MND; central pontine myelinolysis

9. What investigations would you do if you did not know the diagnosis? (3)
   Barium contrast swallow; Upper GI endoscope; Oesophageal manometry

10. What investigations would you do for a bulbar palsy? (1)
    MRI brainstem

Question 10: Hepatitis

A 34-year-old male presents with nausea, arthralgia, hepatosplenomegaly, and jaundice, as well as a fever. He is an IV drug user and is sexually active currently not using protection.

1. What is the diagnosis? (1)
   Hepatitis B/C

2. What are risk factors? (4)
   Men with have sex with other men; Children with have HBsAg + mothers; children adopted from endemic areas; IV drug users; risky sexual behaviour; blood transfusions; health care workers

3. What is the incubation period of Hep B? (1)
   1-6 months

4. What investigations? (5)
   HEP B: HBsAg is present 1-6 months after infection; HBeAg is present for 1-3 months and infers high infectivity. Anti-HBc implies past infection. Anti-HBs only implies vaccination, HBV PCR allows monitoring of response to therapy
   HEP C: Anti-HCV confirms exposure; HCV-PCR confirms chronicity/ infection
   Other: HIV test
   LIVER BIOPSY TO ASSESS DAMAGE AND MALIGNANCY

5. Interpret his lab results: (5)

- ALT 100
- AST 120
- ALP 60
- GGT 45
- Bilirubin 20 mg/dL
- HBsAg positive
- HBeAg positive
- Anti-HBs negative
- Anti-HBc negative
- Anti-HCV negative
- HCV PCR negative

The patient has Hepatocellular damage as indicated by AST, ALT ✅ Bilirubin is high indicating hyperbilirubinemia ✅; The HBsAg is positive indicating a recent infection; the HBeAg is positive indicating the Patient is infectious ✅ The Anti-HBs indicated there is no past infection; anti-HBc is negative indicates there is no vaccination ✅
The Anti-HCV antibodies are negative indicating no Hepatitis C infection ✅. The patient has Hepatitis B.

6. What are the complications? (3)
   Hepatocellular carcinoma; fulminant hepatic failure; cirrhosis; membranous nephropathy , polyarteritis nodosa

7. What is the treatment? (3)
   Immunise sexual partners; stop alcohol; Antivirals (Lamivudine OR Tenofovir OR pegylated interferon)

8. What are some other causes of acute hepatitis besides viral? (4)
   Drugs: Isoniazid; halothane; erythromycin; Acetaminophen (Paracetamol);heavy alcohol intake; chlorpromazine ✅✅✅TOXINS: Amanita phyllodes in mushroom poisoning, carbon tetrachloride✅

Question 11: Diarrhoea

A 24-year-old patient complains of diarrhoea after a party she attended 3 days ago.

1. What is the probable aetiology of the patient? (1)
   gastroenteritis
2. Is it Acute or chronic? What is the cut off? (2)
   Acute <2 weeks
3. What is the commonest organism? (1)
   E.coli, can be Kirbsiella spp. Streptococcus
4. What are 2 other acute causes of diarrhoea? (2)
   Drugs (Metformin, PIs); HIV infection
5. What are 2 chronic causes (2)
   IBD-> UC CD; coeliac disease
6. What if the patient had bloody diarrhoea? What would the infective DD be? (4)
   Shigella, Salmonella; Campylobacter; Amebiasis
7. What would non-infective DD be?(4)
   IBD (UC; Crohn's); Haemorrhoids; Colonic Ca, polyps; Pseudomembranous colitis; ischaemic colitis
8. Give 3 other features that could be present in stool? (3)
   Steatorrhea: Pancreatic insufficiency, biliary obstruction; explosive (Cholera, Yersinia; Giardia); frank pus (IBD); mucus (IBS)
9. What are 3 complications of diarrhoea? (3)
   Dehydration; electrolyte Imbalances; death; hypovolaemic shock
10. 4 signs of dehydration? (4)
    Dec skin turgor; dry mucus membranes; sunken eyes; inc. thirst
11. 5 signs of shock? (4)
    Dec or no UO; dec capillary perfusion rate; hypotension, tachycardia, depressed LoC
12. What are 3 non-GI causes? (3)
    Thyrotoxicosis, Addison's crisis; acute blood loss
13. How would you advise this patient to prevent future diarrhoea episodes? (3)
    hand hygiene; do not eat suspicious foods; always eat cooked food
14. How would you investigate this patient ?(1)
    none treat symptomatically
15. How would you investigate a bloody stool patient? (5)
    FBC; LFT; U&E; ESR, CRP ; US and CT abdomen; colonoscopy and histology

NEURO

Question 1: Dementia

A 70-year-old male presents with loss of memory and his family says he cannot do simple tasks.

1. What is the likely diagnosis?(1)

Dementia

2. What are other causes of the condition (3)

Old age, Infection: HIV, Toxins: Drugs and alcohol, traumatic brain injury, Hyperthyroidism, Deficiencies of folate, Vit B12; Rare: Prion disease

3. What are some reversible causes? (4)

Drugs (antipsychotics or anticholinergics); emotional (depression); metabolic (hypothyroidism, hepatic encephalopathy); electrolyte imbalances (Vit B12; folate deficiency); trauma; tumour or SOL; infection (HIV, syphilis)

4. What is the best screening test? (1)

MMSE

5. Name 4 parts of this screening tool? (4)

Orientation, Registration, Attention & Calculation, Recall, Language, Coping

6. What is a disadvantage of the tool? (1)

The patient would need to be literate/ test is language, education and culture sensitive/ patient has to be able to see and hear/ does not assess executive function

7. What 4 criterion does the patient need to be met in order to confirm diagnosis? (4)

1. The patient needs to have cognitive deficit in 1 or more domains 2. The patient's cognitive deficit interferes with their ability to perform basic tasks 3. The patient has a cognitive deficit independent of delirium 4. The patient's condition cannot be better explained by another neurocognitive condition (schizophrenia, psychosis)

8. What are 4 risk factors? (4)

Alcohol use, drug use, Parkinson's disease, family history, smoking

9. What are the 4 domains affected by the disease? (4)

Learning and memory; language; executive function; perceptual-motor function; complex attention; social cognition function

10. What are 4 differences between the condition and delirium? (4)

Insidious, Chronic progressive course, clear consciousness, patient tries hard to remember (overcome deficit), behaviour often normal, not reversible, normal in early disease, usually in older people, hallucinations are a late feature, word finding difficult

11. What are 3 other types of dementia? (3)

Alzheimer's disease; Lewy Body; Frontotemporal; vascular; Other (Huntington's disease); mixed dementia

Question 2: Seizure

A 15-year-old female presents with convulsions, an aura and tongue biting.

1. What the possible diagnosis? (1)

Seizure

2. What are localising features for specific lobe focal seizures? (4)

Temporal lobe(Automatisms- complex motor phenomena with impaired awareness, primitive oral reflexes(lip smacking, chewing, swallowing) or manual movements (fumbling, fiddling, grabbing) to complex actions, hallucinations of smell, taste or sound ;delusional behaviour ; frontal lobe(motor features, such as posturing or peddling movements of legs; dysphasia or speech arrest, Jacksonian march) ; parietal lobe (sensory disturbances -tingling numbness, pain, motor symptoms due to spread to pre-central gyrus); occipital lobe(visual phenomena such spots, lines, flashes)

3. What are the cause? (5)

Idiopathic, Structural: cortical scarring (head injury years before onset), Developmental (dysembryoplastic neuroepithelial tumour or cortical dysgenesis), SOL, Stroke, vascular malformations; Sarcoidosis; SLE; antibodies to voltage-gates potassium channels; PAN; infections (meningitis, encephalitis)

4. What are post- symptoms? (4) HCW MD

Headache, confusion and myalgia, or temporary weakness after a focal seizure in motor cortex or dysphasia following a focal seizure in the temporal lobe

5. What are provoking causes of seizures? (5) MID WISH

Alcohol and benzodiazepines withdrawal; metabolic/ electrolyte imbalances(hypoxia, hyper/hyponatremia, hypocalcaemia, hyper/hypoglycaemia, uraemia); infection (meningitis, encephalitis); hyperthermia; drugs (tricyclics, cocaine); trauma; stroke; inc. ICP, haemorrhages; neurofibromatosis

6. Treatment ? (1)

All patients with seizure must be referred for specialist assessment and investigation if <2wks.

7. Difference between Focal and generalised seizure.(2)

Focal refers to a specific part/ lobe of the cerebral cortex was affected; generalised refers to a seizure that started in one lobe but moved to other sites

8. Types of focal seizures (2)

Focal seizure with altered consciousness; focal seizures without altered consciousness

9. Types of generalised seizures (3)

Tonic-clonic; myotonic; absence

10. What drug is used to treat seizures? (1)

Phenytoin

11. What drugs are used to abort a seizure ? (2)

Midazolam(IM) & Diazepam (Slow IV)

Question 3: Cerebrovascular accident

a 15-year-old patient presents a left sided hemiplegia with aphasia and presents with left sided facial weakness.

1. What is the diagnosis? (1)

Cerebrovascular accident

2. What are the two types? (2)

Haemorrhagic and ischaemic

3. What is the aetiology of a young stroke? (5)

Vasculitis; SLE; SAH; venous sinus thrombosis; AF; Head trauma; antiphospholipid syndrome; Thrombophilia(Factor 5 Leiden mutation; deficiency in protein C and S; Antithrombin III deficiency

4. What part of the system is affected? (2)

Right ✅cerebral cortex✅ most likely MCA

5. What is the specific, territory affected if the patient tells you the weakness is worse in the upper limb than lower limb? (1)

Territory of brain supplied by Middle cerebral artery (internal capsule)

6. What cranial nerve is affected? (1)

CN7 facial nerve

7. What is the signs of LMN vs UMN lesions of the CN? (3)

UMN -> contra lateral facial weakness; lower part of face usually affected; drooping of mouth; loss of nasolabial fold, sparing of forehead.

LMN-> ipsilateral facial weakness; upper and lower part of face; loss of forehead creases; Bell's phenomena (eye turn up asked to close eyes); cannot raise affected side forehead

8. If the patient had a UMN lesion, what are 5 signs you would see? (5)

Hyperreflexia; hypertonic; diminished power; slight/ no wasting of muscles; Upgoing plantar reflex

Question 4: Meningitis

a 24-year-old male , HIV reactive presents with neck stiffness and headache. His VL is very high and CD4 count is 2.

1. What is the possible diagnosis? (1)

Meningitis

2. Explain 3 ways HIV affects the CNS? (3)

Direct effect on neuroglial cells(HIV dementia; HIV neuropathy; Chronic asymptomatic meningitis); Immune dysregulation (IDP; aseptic meningitis; Myasthenia Gravis; polymyositis); HIV DRUGS -EFV: Neuropsychiatric effects ; INH: neuropathy; AZT: myopathy; IRIS

3. What are 4 other symptoms the patient might have ?(4)

Photophobia, nausea, and vomiting; depressed level of consciousness; drowsiness; confusion; rash

4. What is the possible pathogens if this is the acute bacterial form? (4)

Streptococcus pneumoniae; Neisseria meningitides; staphylococcus aureus; haemophilia influenzae; listeria monocytogenes

5. What are 5 non- infectious causes? (5)

Non-inflammatory: SLE, RA, Sjogren's syndrome; Bechet' syndrome ; Sub-arachnoid haemorrhage; Malignancy(primary or secondary); drugs (IT Methotrexate; Isoniazid; Sulfonimines; NSAIDS)

6. What is the diagnostic test?(1)

LP

7. What signs could the patient have ? (2)

Positive Kernig's and Brudzinski signs

8. What are some C/Is to the above test ?(4)

Raised Intra-cranial pressure; SOL; haemophilia; infection over the site; patient refusal

9. The patient is found to have acute bacterial meningitis, what abnormalities would the CSF have? (5)

Bacterial meningitis:

Appearance: Turbid, yellow ✅

Glucose: low <2 mmol/L ✅

Protein: high >2.2 g/L ✅

WCC: high >2000

Predominant cell: Neutrophils >1180 ✅

Culture: Meningococcus, pneumococcus; haemophilus ✅

10. What are 3 complications of the condition? (3)‼‼

Cerebral abscess; Cranial nerves palsy; hydrocephalus; dehydration

Question 5: Subarachnoid Haemorrhage

A 27-year-old male comes in with an excruciating headache after exercising at the gym. He has some neck stiffness as well and is in a lot of pain.

1. What is the likely diagnosis? (1)

Subarachnoid Haemorrhage

2. What other signs might be present?(3)

Kernig's sign(takes 6 hrs); focal neurology such as pupil changes could indicate 3rd CN palsy with posterior communicating artery aneurysm; retinal and vitreous bleeds

3. What are 3 causes? (3)

Berry aneurysm rupture of circle of Willis; AV malformations; Encephalitis; Vasculitis; Tumour; Idiopathic

4. What are 3 risk factors ?(3)

Hypertension; thrombophilia; smoking ; alcohol use; family history; previous aneurysmal SAH; Polycystic kidney disease; aortic coarctation and Ehlers's Danlos's syndrome

5. Differentials ?(3)

Meningitis, migraine ; intracerebral bleed; dissection of a carotid of vertebral artery; benign thunderclap headache(triggered by Valsalva manoeuvre e.g., cough, coitus)

6. Investigations? (2)

Urgent Head CT-> detects >95% of SAH within 1st 24 hr

Consider LP if CT negative needs to be done >12 hr after headache onset

7. Management(3)

Refer!!! Re-examine CNS often, chart BP, pupils, and GCS; maintain cerebral perfusion by keeping hydrated; Nimodipine is a Ca antagonist that reduces vasospasm and consequent morbidity from cerebral ischaemia; Surgery: endovascular coiling or surgical clipping

8. Complications(4)

Rebleeding; Hydrocephalus; Cerebral ischaemia; Hyponatraemia

Random internal medicine questions:

Respiratory:

1. Respiratory causes of clubbing
   Lung cancer, bronchiectasis, lung abscess, cystic/pulmonary fibrosis, empyema
2. Non respiratory causes of clubbing
   CVS: Infective endocarditis, cyanotic congenital heart disease
   GIT: IBS, Cirrhosis, GI Lymphoma
3. Causes of exudate
   Community acquired pneumonia, TB, embolism, RA, SLE, carcinoma
4. Causes of transudate
   RHF, Renal failure, chronic liver disease, Nephrotic syndrome
5. Causes of tracheal deviation towards lesion
   Collapse, fibrosis
6. Causes of tracheal deviation away from lesion
   Pleural effusion, pneumothorax
7. Turbid pleural effusion liquid
   Empyema
8. Coarse vs light crackles
   Light = pneumonia
   Coarse = bronchiectasis
9. 2 TB Tests
   Gene Xpert, Mantoux (tuberculin skin test)
10. 2 corticosteroids to use in asthma
    Beclomethasone, Prednisolone
11. Example of SABA
    Salbutamol
12. Example of LABA
    Salmeterol
13. Distinguish between type 1 and type 2 respiratory failure
    Type 1 = hypoxia = dyspnoea, restless, agitation, confusion, central cyanosis
    Type 2 = hypercapnia = headache, tachycardia, bounding pulse, tremor, drowsy, coma, peripheral vasodilation
14. 4 Signs of Respiratory distress
    Tachypnoea, nasal flaring, tracheal tug, use of accessory muscles, intercostal/subcostal recessions, pulsus paradoxes
15. Causes of respiratory failure
    Asthma, COPD, Pneumonia, pulmonary oedema, pulmonary embolism, ARDS
16. List 2 organisms that cause pneumonia
    Streptococcus pneumonia, H. influenzae
17. List 2 viruses that cause pneumonia
    Influenza, cytomegalovirus
18. Grade the dyspnoea using MRC Grading
    1: exercise
    2: walking up
    3: walking flat
    4: walking slow
    5: can't leave home
19. Causes of bronchiectasis
    Congenital: cystic fibrosis, young Kartagener's
    Acquired: TB, tumour, pneumonia (same organisms)
20. 5 signs of Bronchial carcinoma
    Cough, dysphonia, dyspnoea, chest pain, clubbing, pallor, wasting, lymphadenopathy(supraclavicular and axillary)

CVS:

1. 4 signs and symptoms of Left Heart failure
   Dyspnoea, fatigue, orthopnoea, wheeze, paroxysmal nocturnal dyspnoea, nocturnal cough, cold peripheries
2. 4 signs and symptoms of right heart failure
   Peripheral oedema, ascites, facial engorgement, nausea, JVP
3. 4 drugs classes in the management of heart failure
   Diuretics, Ace-I, B-Blocker, Vasodilators, and digoxin (drug)
4. New York classification of heart failure using dyspnoea
   1: HD, no dyspnoea
   2: comfort at rest, dyspnoea on heavy activity
   3: dyspnoea on activity
   4: dyspnoea at rest
5. RHD pathophysiology
   Heart valves are permanently damaged by RF due to thickening of leaflets, thickening/fusion of tendinous cords and commissural fusion and shortening
6. Jones major and minor criteria
   Major: CASES: carditis, arthritis, subcutaneous nodules, erythema marginatum, Sydenham's chorea
   Minor: FEAPP: Fever, ↑ ESR, arthralgia, prolonged PR interval, previous Rheumatic fever
7. What organism causes RHD
   Group A B-Haemolytic streptococcus
8. Treatment of choice for RHD
   Penicillin (benzylpenicillin)
9. Causes of infection endocarditis
   Strep. Viridans (most common), strep. Aureus
   Risk factors: RHD, IV drug users, cardiac lesions
10. Duke's major and minor criteria
    Major: 2 positive blood cultures, endocardial involvement
    Minor: Fever, IV Drugs, Osler nodes/Roth spots, Janeway lesions, echo findings
11. Firstline management of Infective endocarditis
    Benzylpenicillin, gentamycin
12. Risk factors for myocardial infarction
    Non modifiable: Age, male, family history of IHD
    Modifiable: smoking, hypertension, DM, hyperlipidaemia, obesity, cocaine
13. Types of Coronary Artery disease (CAD)
    Unstable angina, STEMI, NSTEM
14. Common ECG findings in STEMI
    ST elevation, inverted T waves
15. 2 signs in Atrial Fibrillation
    Irregularly irregular pulse, rapid heart rate
16. ECG findings in atrial fibrillation
    Absent P waves, irregular R intervals, narrow QRS complex
17. Firstline treatment for Hypertension
    ACE-I = enalapril or Thiazide diuretic = HCTZ
18. Signs of angina
    Pain worse with swallowing, associated with palpitations, relieved by rest, brought on by exertion, emotion, cold weather, heavy meals
19. ECG findings in angina
    ST Depression with flat or inverted T waves
20. Apex beat character in aortic stenosis
    Heaving = pressure loaded
21. Mitral regurgitation descriptive findings
    Pansystolic, mitral area radiating to axilla, louder on expiration
22. Pulse in Aortic regurgitation
    Parvis – Tardis
23. Pulse in mitral stenosis
    Irregularly irregular, low volume (A Fib)

GIT

1. 2 Causes of acute liver disease
   Alcohol, trauma, drugs (Tylenol), Hep B
2. 4 signs of acute liver disease
   Jaundice, asterixis (hepatic flap), hepatic encephalopathy, ascites, nausea, vomiting
3. What are 3 things you would look for in the investigations
   Increased prothrombin time, Elevated aminotransferase, elevated bilirubin
4. 2 causes of chronic liver disease
   Chronic alcohol use, HBV/HCV infection, Budd Chiari
5. 4 signs of chronic liver disease
   Leukonychia, clubbing, terry's nails, palmar erythema, Dupuytren's contracture, hepatosplenomegaly, spider naevi, ascites, xanthelasma
6. 2 causes of clubbing as a result from GIT disease
   IBS, Cirrhosis, GI lymphoma
7. Name 4 investigation that you would do in chronic liver disease
   Bedside glucose, LFT, FBC, U&E, Liver ultrasound
8. 2 complications of chronic liver disease
   Portal hypertension, liver failure, ascites, malignant transformation
9. 3 signs of IBD
   Abdominal discomfort with defecation relief, altered stool, altered frequency, urgency, tenderness
10. Investigations to note in Ascites
    Abdominal ultrasound, diagnostic paracentesis, Serum albumin ascites gradient, glucose, shifting dullness
11. Management of ascites
    Sodium restriction and diuretics (spironolactone/furosemide)
12. What are the ALARM symptoms in Peptic ulcer disease
    Anaemia, LOW, Anorexia, recent onset, melaena, swallowing difficulty
13. What is the most common organism that causes peptic ulcer disease
    Helicobacter pylori
14. 4 factors contributing to Gastroesophageal reflux disease
    Smoking, alcohol, pregnancy, gastric acid hypersecretion, oesophageal dysmotility
15. 4 signs of GERD
    Heartburn, belching, acid brash, chronic cough, laryngitis
16. 2 complications of GERD
    Oesophagitis, ulcers
17. What is the non-pharmacological management of GERD
    Weight loss, smoking cessation, smaller meals, avoid caffeine, hot food, chocolate, stop eating 3 hours before sleeping and raise the head of the bed
18. 4 signs of chronic pancreatitis
    Epigastric pain relieved by sitting forward, bloating, steatorrhea, LOW, grey turners, Cullen's
19. Name 3 investigations you would do if you suspected Chronic pancreatitis
    Elevated pancreatic enzymes, raised lipase, Hyperamylasuira
20. What organism causes bloody diarrhoea
    Shigella, salmonella, e. coli
21. What endoscopy procedure would you do
    Colonoscopy
22. List 2 stool investigations which you could perform
    Stool MC&S and Faecal occult blood test (FOBT)

Neuro:

1. Name 5 red flags to note in a headache
   SNOOP: SAH/Systemic, neurological symptoms, onset, older patients, previous history
2. 3 symptoms of a headache
   Pain, nausea, vomiting, photophobia, allodynia, dysphagia, dysarthria
3. 3 triggers of a headache
   Alcohol, travel, exercise, oral contraceptives, caffeine
4. Name the 2 types of a stroke
   Ischaemic and haemorrhagic
5. Risk factors for a stroke
   Modifiable: HPT, DM, obesity, dyslipidaemia, poor diet
   Non modifiable: Male, increased age
6. Localise the lesion of a stroke
   Arm > leg = MCA
   Leg = arm = ACA
   (Hemiplegia often in internal capsule)
7. 2 bacterial and 2 viral causes of meningitis
   Bacterial: Strep pneumoniae, Neisseria meningitidis, TB
   Viral: Coxsackievirus, HSV, HIV
   Fungal = CCM (cryptococcal), candida albicans
8. 4 signs of meningism
   Brudzinski, Kernig's, neck stiffness, photophobia, seizures, vomiting, confusion, Cushing's triad (hypertension, bradycardia, irregular breathing)
9. CSF interpretation of bacterial, viral and TB meningitis
   Bacterial = Turbid, neutrophils, high protein, low glucose
   Viral = clear, leucocytes, high protein
   TB = straw, leucocytes, high protein, low glucose
10. 3 symptoms of delirium
    Impaired cognition/perception, consciousness over hours/days with marked memory deficit and disorientated thinking
11. 2 types of dementia
    Alzheimer's, vascular Lewy bodies
12. Reversible causes of dementia
    Depression, vit b12 deficiency, diabetes, thyroid problems, infection
13. What is the classic triad in Parkinson's
    Pin rolling resting tremor, hypertonia, bradykinesia
14. What is the first line therapy in Parkinson's disease
    Levodopa
15. Differentiate between the types of seizures
    Focal: unilateral with/without conscious impairment
    Generalised: absent, tonic-clonic, myoclonic, atonic
16. Different triggers of seizures
    Trauma, hypoglycaemia, hypoxia, alcohol, cocaine, tramadol
17. First line management of seizure
    Carbamazepine or sodium valproate
18. 5 signs of Myasthenia gravis
    Myasthenic snarl, ptosis, curtain sign, peak sign, voice fatiguability, facial muscle fatigue
19. 2 investigations that you would look for to confirm your diagnosis of myasthenia gravis
    Serum antibodies: Anti-AchR and MUSK (muscle specific tyrosine kinase)
20. 3 signs of multiple sclerosis
    Sensory loss, spastic weakness, unilateral optic neuritis, swallowing disorder, trunk and limb ataxia, cognitive decline

Rheumatology:

1. Risk factors for RA
   Twins, human leucocyte antigen, smoking hormones
2. 4 signs of RA
   Symmetrical swollen joints worse in morning, inflammation, bursitis, ulnar deviation, swan neck deformity, boutonniere deformity
3. What would you note on investigation for RA
   Positive Rheumatoid factor and anticyclic citrullinated peptide antibodies
4. What are the extra-articular manifestations of RA
   Skin: Subcutaneous nodules, vasculitis
   Eyes: keratoconjunctivitis, scleritis
   Lungs: pleural effusion, interstitial lung disease
   heart: pericarditis/effusion
   feet: hallux valgus, hammer toe
5. 4 causes of osteoarthritis
   Elderly, family history, collagen 2A gene, obesity
6. Which joints are most commonly affected by osteoarthritis
   Knee and hip (pain and crepitus on movement, ache at rest)
7. What would you note on the radiological examination of osteoarthritis
   LOSS: loss of joint space, osteophytes, subarticular sclerosis, subchondral cysts
8. 4 risk factors for gout
   Obesity, alcohol, high protein diet, HPT, DM, dyslipidaemia
9. What would you notice in the tendons
   Tophi and deposits of urate crystals
10. Name 3 non-pharmacological treatment methods to Gout
    Weight loss, rest, avoid fasting, alcohol, purine rich meat
11. 5 symptoms of SLE
    Butterfly rash, discoid rash, non-scarring alopecia, oral ulcers, Raynaud's phenomenon
12. 6 investigations to do if you suspect SLE
    +ANA, anti-dsDNA, Anti-smith antibodies, antiphospholipid abs, low complement c3/c4, + direct coombs test
13. How would you council a patient with SLE
    High factor sunblock, hydroxychloroquine, screen for diseases

Haematology:

1. What are the causes of iron deficiency anaemia
   Blood loss, poor diet, malabsorption
2. What investigations would you do in iron deficiency anaemia
   FBC, Iron studies, smear
3. What is the definitive management of iron deficiency anaemia
   Iron supplements = ferrous sulfate 200mg
4. What are important questions to ask on history for iron deficiency anaemia
   Diet, menorrhagia
5. What are the causes of b12/folate deficiency anaemia
   Reduced thiamine, reduced RBC production, vegans, tapeworms, gastrectomy
6. 4 signs on examination of b12/folate deficiency anaemia
   Yellow tinge, anaemic features (pallor, fatigue, headache), Babinski with absent knee/ankle jerk
7. 4 things to note on investigation with b12/folate deficiency
   Degreased Hb, decreased MCV, decreased serum b12, decreased platelets, Hypersegmented neutrophils

Renal

1. Staging of acute renal injury
   (Looks at creatine and urine output)
   1: C=1.5-1.9(>26)/ U=<0.5/6h
   2: C=2-2.9 / U=<0,5/12h
   3: C=3x/ U=<0.3/24h
2. 3 signs of acute renal injury
   Small kidneys, fluid overload, arrythmias
3. 3 causes of acute renal injury
   Pre: decreased circulating volume, CCF
   Renal: glomerulonephritis, acute tubular necrosis, hypertensive emergencies
   Post: bladder/urethral obstruction
4. Staging of chronic renal injury
   Grading: GFR
   1: >90 (no evidence)
   2: 60-89
   3a: 45-59
   3b: 30-44
   4: 15-29
   5:<15 = ESRD
5. 4 causes of chronic renal injury
   HPT, DM, HIV, pyelonephritis, NSAIDS
6. 4 signs in chronic renal injury
   Fluid overload, nausea, vomiting, nocturia, polyuria, bruising, dark foamy urine
7. Signs of nephrotic syndrome
   PHO: proteinuria, hypoalbuminemia, oedema (periorbital)
8. Causes of nephrotic syndrome
   Diabetic nephropathy, minimal change disease (rifampicin), focal segmental glomerulosclerosis
9. Signs of nephritic syndrome
   CHOC: cola urine, hypertension, oliguria, haematuria
10. Causes of nephric syndrome
    SLE, IgA nephropathy, group A B haem strep

Endocrine:

1. Differentiate between type 1 and type 2 diabetes
   T1: insulin deficiency, child, LOW
   T2: insulin resistance, adult, BMI
2. Firstline treatment in type 1
   Insulin
3. Firstline treatment in type 2
   Metformin
4. Name 4 investigations to do in diabetes
   Random glucose, fasting glucose, HbA1C, OGTT (post load glucose)
5. 4 signs of DKA
   Ketonic breath, air hunger, weakness, abdominal pain, poor vision, polyuria
6. 4 signs of HHS
   Tachycardia, hypotension, dehydration, confusion, coma, Kussmaul breathing
7. Management of diabetic emergency
   Insulin but beware of hypoglycaemia, give bicarbonate if acidotic
8. 4 signs of hyperthyroidism
   LOW, diarrhoea, heat intolerance, sweaty palms, clubbing, Plummers nails, Bi lobe goitre with bruit
9. Treatment of hyperthyroidism
   Anti-thyroid drug, subtotal thyroidectomy if large goitre
10. 4 signs of hypothyroidism
    Weight gain, cold intolerance, cold peripheries, constipation, puffy face, poor memory, bradycardia, carpal tunnel syndrome
11. Treatment of hypothyroidism
    T4 = levothyroxine
12. 3 signs of Cushing's
    Weight gain, obesity, moon face, bruises, hypertension, proximal weakness, mood changes
13. 3 signs of Addison's
    Lean, tanned, tired, depression, nausea, vomiting

Infectious diseases:

1. 5 signs of TB
   Fever, LOW, night sweats, malaise, anorexia, clubbing
2. 3 investigations to do if you suspect TB
   Mantoux, interferon-gamma release assays, Gene Xpert, nucleic acid amplification test
3. Management of TB
   Rifafour = rifampicin, isoniazid, pyrazimide, ethambutol
4. Staging of HIV according to the WHO
   Stage 1: Primary infection, asymptomatic, generalised lymphadenopathy
   Stage 2: recurrent URTI, weight loss <10%, mucocutaneous manifestations, herpes zoster
   Stage 3: Prolonged fever/diarrhoea, oral candidiasis, weight>10%, severe bacterial infection
   Stage 4: AIDS-defining conditions = pneumocystis pneumoniae
5. 4 test that you would do to confirm HIV
   ELISA, viral PCR, rapid, CD4, Viral load
6. 3 classes of HIV drugs
   NRTI'S (nucleoside reverse transcriptase inhibitors) , NNRTI'S, protease inhibitors
7. Go to drug combination for HIV positive patients
   Atripla = lamivudine, tenofovir, efavirenz
8. What organism cases syphilis
   Treponema pallidum
9. Differentiate between the staging of syphilis
   Primary:<90days, macule-papule-chancre
   Secondary:4-10weeks after chancre, rash, condyloma Lata
   Tertiary= 20-40years Gumma
10. Definitive management of syphilis
    PRC, RPR, Penicillin (Benzathine penicillin)